



Contents lists available at ScienceDirect
Waste Management Bulletin

journal homepage: www.elsevier.com/locate/wmb



Assessment of the challenges to implementing green manufacturing in the food and beverage industry: an emerging economy perspective

Maishat Lubaba^a, Sanjida Hassan^b, Munera Sultana^b, Shah Murtoza Morshed^b, A.B.M. Mainul Bari^{b,*}

^a Department of Mechanical and Production Engineering, Ahsanullah University of Science and Technology, Dhaka 1208, Bangladesh ^b

Department of Industrial and Production Engineering, Bangladesh University of Engineering and Technology, Dhaka 1000, Bangladesh

ARTICLE INFO

Keywords:

Green manufacturing
Waste management
Food and beverage industry
Bayesian Best-Worst Method
Probabilistic group decision-making
Sustainability
Emerging economy

ABSTRACT

Green Manufacturing (GM) is a pivotal strategy for fostering sustainability in the food and beverage industry, particularly in emerging economies. It addresses critical environmental issues, including high carbon emissions, excessive waste, and water pollution, while improving resource efficiency and aligning industrial practices with global environmental standards. Despite its potential benefits, the implementation of GM in the food and beverage sector faces significant, often underexplored, challenges, particularly in resource-constrained emerging economies. These obstacles hinder effective decision-making and the adoption of sustainable practices. This study aims to explore the challenges associated with GM implementation in the food and beverage industry within the context of an emerging economy like Bangladesh. Through a comprehensive literature review and expert consultations, seventeen key challenges were identified. These challenges were evaluated using the Bayesian Best-Worst Method (BWM), which is a probabilistic group decision-making approach. The findings reveal that the top three challenges are: "Difficulty to incorporate eco-friendly design concepts" (weight: 0.0740), "Absence of efficient recycling facilities" (weight: 0.0658), and "High initial investment cost and lack of financial support" (weight: 0.0657). The study provides a strategic framework for policymakers and industry leaders to address these barriers effectively. It also outlines actionable guidelines and strategies to facilitate the transition toward sustainable practices, ensuring environmental, economic, and social sustainability while enhancing industrial competitiveness and performance.

Introduction

Rapid industrialization and urbanization have significantly depleted natural resources, exacerbating global resource scarcity (Karuppiyah et al., 2020). Key challenges such as limited natural resources, air quality degradation, and inadequate wastewater management threaten economic growth (Sharma et al., 2021). Notably, the food-processing industry, characterized by low-profit margins (Mason et al., 2023), contributes to environmental degradation through resource-intensive practices and wastewater exposure (Singh et al., 2023), necessitating the adoption of resource-efficient measures (Michel et al., 2024), renewable energy integration (Savvakis et al., 2023), and improved productivity (Akanmu and Nordin, 2022; Chen and Voigt, 2020). This highlights the urgent need for sustainable practices in this sector to mitigate environmental and economic risks (Hariram et al., 2023). Therefore, industrialization strains resources and necessitates sustainability in sectors like food processing to address environmental and economic challenges.

Sustainability has gained prominence as environmentally conscious consumers drive industries toward adopting eco-

friendly practices (Ullah et al., 2021). Growing societal awareness has encouraged the industrial sectors to show greater commitment toward sustainable development (Karuppiyah et al., 2023). Many developed countries have successfully implemented "Green Manufacturing" (GM) in large-scale industries, emphasizing manufacturing techniques that minimize environmental impact (Bendig et al., 2023). However, developing nations face barriers such as outdated machinery (Atieh et al., 2023), limited innovation (Mer and Virdi, 2024), unskilled labor (Alam et al., 2024), and insufficient organizational collaboration (Karuppiyah et al., 2020; Adam et al., 2020). For developing countries across the globe, striking a balance between industrial progress and environmental sustainability remains a persistent challenge (Zheng et al., 2025). Small-scale industries, including the food and beverage sectors, remain overlooked in favor of high-tech and energy-intensive industries, further hindered by inadequate capital and lack of government incentives (Chien et al., 2021). Economic constraints in emerging economies prioritize survival over innovation, leaving GM adoption in a nascent stage (Premadasa et al., 2023). While GM adoption progresses in developed economies (D'Angelo et al., 2023), emerging

* Corresponding author.

E-mail address: mainul.ipe@gmail.com (A.B.M.M. Bari).

economies face structural and financial barriers, especially in small-scale industries like food and beverages (Agarwal et al., 2024). Despite these challenges, manufacturing organizations are increasingly pressured to integrate sustainability into their production system (Karuppiyah et al., 2024).

Advocates for green technology emphasize its potential to align environmental performance with economic benefits, urging industries to transition from environmental management to proactive strategies (Saxena and Srivastava, 2022). Traditional manufacturing is marred by inefficiencies such as excessive energy consumption, pollution, and limited recycling opportunities (Li et al., 2022). GM, through eco-friendly production methods, minimal resource utilization, and renewable energy adoption, offers a sustainable alternative (Yang et al., 2022). For instance, co-locating food manufacturing facilities with biorefineries can optimize food waste management and logistics (Sovacool et al., 2021). In contrast to traditional practices, GM fosters recycling, reduces waste (Vangeri et al., 2024), and enhances sustainability (Rajpoot, 2024), meeting consumer demands for eco-friendly products (Khang and Akhai, 2024). Thus, GM enhances sustainability and economic efficiency by addressing inefficiencies in traditional manufacturing and meeting consumer demand for eco-conscious practices.

Fig. 1 highlights the key components of GM and their potential applications in the food and beverage industry, emphasizing a comprehensive approach to sustainability. Water conservation is another critical aspect of GM, involving efficient systems to optimize water usage and limit wastewater generation, which is particularly important for resource-intensive industries like food and beverage (Das et al., 2023). Moreover, sustainable packaging practices, including the use of biodegradable, compostable, or recyclable materials, address environmental concerns related to packaging waste (Sabet, 2024). Decarbonization technologies, such as carbon capture, storage, and utilization, are essential to reducing greenhouse gas emissions (Nagaj et al., 2024). Smart manufacturing leverages IoT and AI to enhance resource efficiency, monitor production, and minimize waste (Tyagi et al.,

which are renewable, non-toxic, and sustainable, helps reduce environmental harm (Renita et al., 2023). GM also needs the integration of renewable energy sources, such as solar and wind, to reduce reliance on fossil fuels and minimize carbon emissions (Usman et al., 2024). Lastly, waste management strategies, including recycling, upcycling, and waste-to-energy initiatives, are implemented to reduce landfill dependency and promote resource recovery (Shah et al., 2024). Together, these interconnected strategies form a holistic framework for GM, enabling the food and beverage industry to achieve environmental, economic, and social sustainability while meeting consumer expectations and regulatory demands.

Achieving sustainability in GM necessitates integrating economic, environmental, and social dimensions, the triple bottom line (TBL) approach. TBL offers strategic benefits such as energy efficiency, reduced waste, improved decision-making, and workforce well-being (Ali et al., 2023; Khan et al., 2021). While developing economies recognize these advantages, weak regulatory frameworks have failed to curb environmental pollution, underscoring the need for industry-driven GM initiatives (An et al., 2021). GM processes, including carbon emission reduction, eco-friendly energy adoption, and recycling, can mitigate environmental harm while supporting economic growth (Kwakwa, 2022). Furthermore, fostering green behaviors and sustainable human resource practices contributes to social sustainability (Amrutha and Geetha, 2020). Therefore, TBL in GM integrates environmental, economic, and social goals, offering strategic benefits and addressing regulatory gaps in developing economies.

Memon et al. (2021) consider the food processing industry one of the fastest-expanding industries because of its characteristics as a dynamic and diverse business area. This sector faces compounding challenges from events like the COVID-19 pandemic and geopolitical disruptions like the Russia-Ukraine conflict (Sommerville, 2024). These crises underscore the urgency to transition to sustainable practices, with GM emerging as a viable solution to energy and resource constraints (Chowdhury et



Fig. 1. Key components of green manufacturing (GM) and their applications. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2024). Eco-design focuses on sustainability by utilizing biodegradable and recyclable materials in product development (Singhal et al., 2024). Additionally, the use of green materials,

al., 2022; Jagtap et al., 2022). The post-pandemic era presents an opportunity for emerging economies to adopt GM as a first-mover advantage, promoting innovation and sustainability

(Mohideen et al., 2021; Liu and Yi, 2017). Some previous studies on Bangladesh's food and beverage sector have explored sustainability, but their primary focus was on technological advancement. For instance, Jerin et al. (2025) investigated the factors influencing food and beverage SMEs' intention to adopt advanced technologies, while Rahman et al. (2025) identified barriers to Industry 4.0 adoption to accelerate its implementation in the sector. According to the preceding discourse, it's evident that there is a lack of research on GM adoption in the food and beverage industry in the context of emerging economies. Therefore, this study focuses on the following research questions (RQs):

RQ1: What are the challenges to adopting GM in the food and beverage industry?

RQ2: How can these challenges be evaluated, and what is the hierarchical ranking among the identified challenges?

RQ3: How can addressing these challenges promote sustainability in the food and beverage industry?

To meet the above RQs, the corresponding research objectives (ROs) are:

RO1: To identify challenges for GM implementation in the food and beverage industry.

RO2: To evaluate and prioritize the identified challenges hierarchically.

RO3: To assist policymakers and managers in formulating action plans for GM adoption in the food and beverage industry to enhance sustainability.

To fulfill these above-mentioned ROs, the identified challenges have been evaluated and prioritized by employing a structured probabilistic group decision-making framework, the Bayesian Best-Worst Method (BWM). Bayesian BWM is an advanced multi-criteria decision-making (MCDM) technique. Where the traditional BWM relies solely on the opinion of one expert to determine how to weigh a number of criteria (Kizielewicz et al., 2024), the Bayesian BWM integrates multiple experts' opinions using statistical estimation methods (Imran et al., 2024). In a Bayesian perspective, aggregated weight shows the degree to which one criterion is favored over another based on the preferences of all experts (Munim et al., 2022). Bayesian BWM has been chosen for this study as it offers a probabilistic approach to solving problems, allowing a group of professionals to be involved, and helps to achieve consensus among these professionals' viewpoints without any information loss (Hossain et al., 2024). This advanced analytical technique allows for a more adaptable and precise understanding of GM implementation challenges in the food and beverage industry. In comparison with other traditional MCDM techniques such as AHP, TOPSIS, or VIKOR, it has a stronger capability to manage uncertainty and complexity that offers a deeper and more realistic evaluation (Mohammadi and Rezaei, 2020; Imran et al., 2024). In the context of Bangladesh, where expert time and consistent data are constrained, Bayesian BWM can be both an efficient and a reliable choice. It minimizes the number of pairwise comparisons required from experts. This eases their cognitive effort but enhances the consistency of results. Credal ranking in Bayesian BWM assists in interpreting the order of significance of each criterion (Saner et al., 2022). Furthermore, traditional fuzzy methods rely heavily on subjective membership functions and are sensitive to inter-rater agreement, often leading to information loss or ambiguity when consensus is low. In contrast, Bayesian BWM captures individual expert uncertainty and diversity through posterior distributions without forcing convergence, thus retaining granular decision dynamics and preserving the credibility of minority expert opinions. Overall, this method supports both theoretical

advancement and practical insights, which deliver well-grounded recommendations for overcoming GM challenges.

Related studies, research gaps, and study contribution

Green manufacturing (GM) methods enable companies to achieve enhanced success (Wang et al., 2023), competitiveness (Queiroz et al., 2024), and profitability by optimizing processes (Vangeri et al., 2024), reducing waste (Bapat et al., 2024), and improving performance across environmental, operational, and economic dimensions (Haleem et al., 2023; D'Angelo et al., 2023). GM addresses critical manufacturing challenges such as environmental compliance (Guo et al., 2024), resource conservation (Guan et al., 2023), hazardous substance control (Awewomom et al., 2024), and waste management (Kuppan et al., 2024). It helps firms redesign their operational systems to comply with stringent environmental regulations while preserving the natural environment (Nazir et al., 2024). In the food and beverage sector, major challenges like food waste, production losses, and plastic pollution significantly affect sustainability, as highlighted by Ni'zeti'c et al. (2019). Green technological innovations improve resource utilization efficiency and mitigate pollutant emissions during production (Feng et al., 2021). Furthermore, adopting recyclable practices (Kurniawan et al., 2023), such as waste recovery (Phonthanukitithaworn et al., 2024), recycling (Maldonado-Romo et al., 2024), and remanufacturing (Zorpas, 2024), presents effective and sustainable solutions for managing plastic waste (Alaghemandi, 2024).

The adoption of GM plays a vital role in achieving environmental sustainability (Umar et al., 2024), especially as governments across the globe enforce stricter regulations to minimize ecological damage and promote better environmental performance (Cheng et al., 2024). However, emerging economies face significant challenges such as limited knowledge among top management (Fahad et al., 2022), lack of skilled workforce (Karmaker et al., 2023), and the absence of dedicated research and development (R&D) teams (Nguyen et al., 2022), which hinder the adoption of GM practices. These barriers underscore the pressing need to address capability-building in organizational structures for GM implementation. Despite its benefits, GM adoption faces structural and knowledge-related barriers in emerging economies, necessitating strategic interventions.

Existing literature extensively evaluates the drivers and barriers to GM adoption. For example, Shukla et al. (2023) examined barriers and drivers in Indian manufacturing industries through a Likert-scale survey. However, their reliance on arithmetic mean calculations might lead to information loss. Karuppiiah et al. (2020) explored GM barriers using Decision-Making Trial and Evaluation Laboratory (DEMATEL), Analytical Network Process (ANP), and TOPSIS in a fuzzy context, identifying internal strategies and lack of eco-design as critical issues. For small and medium enterprises (SMEs) in India, limited R&D and inadequate accreditation remain significant obstacles. Zain et al. (2024) examined the challenges and drivers influencing the adoption of Green Supply Chain Management (GSCM) in Malaysia's food and beverage SMEs. They identified a lack of knowledge and information sharing as the main barrier and highlighted green manufacturing as the most significant driver in promoting sustainable practices. Various methodologies have been applied to identify GM barriers, but limitations in data integration and industry-specific nuances remain underexplored.

Huang et al. (2022) assessed barriers to green innovation in Vietnam's manufacturing sector using Interpretive Structural Modeling (ISM) and MICMAC analysis, highlighting financial constraints and insufficient government support as significant obstacles. Escobar et al. (2023) conducted a bibliometric analysis

using VOSviewer, incorporating co-occurrence networks and cluster analysis to identify key research themes and influential literature on barriers to sustainable green innovation, while Shahin et al. (2024) employed the Intuitionistic Fuzzy Multiplicative Best-Worst Method (IFMBWM) to determine the weights of barriers to green innovation in SMEs, and applied the Interval-Valued Intuitionistic Fuzzy MULTIMOORA (IVIF-MULTIMOORA) method to rank the strategies to overcome these barriers. Runtuk et al. (2024) analyzed focus group discussions to identify challenges and solutions in integrating green suppliers in manufacturing, where the key issues identified were high costs, limited materials, and quality verification, and solutions involved strategic partnerships, audits, and clear evaluation criteria to enhance sustainability and competitiveness. Moreover, Gull et al. (2024) demonstrated that top management's green commitment and an adaptability culture positively influence organizational green performance, and this relationship is mediated by green product innovation. However, previous studies on GM barriers have primarily focused on isolated perspectives, lacking an integrated approach to prioritize challenges effectively.

Table A1 of Appendix A in the *supplementary materials* summarizes recent studies employing Bayesian BWM in diverse industries and geographic contexts, highlighting its utility for evaluating critical factors and prioritizing strategies. Key insights from these studies emphasize the method's adaptability in identifying and ranking challenges, risks, and success factors in various domains. Applied predominantly in emerging economies and industrial sectors such as furniture, steel, textile, footwear, and agriculture, Bayesian BWM consistently provides precise rankings to aid decision-making. Thus, it's evident that Bayesian BWM is ideal for ranking multifaceted barriers such as high costs, infrastructure constraints, and consumer behavior for GM adoption in the food and beverage industry. Fig. S1

Despite these advancements, no research has specifically assessed GM barriers in the food and beverage sector using the Bayesian Best Worst Method (BWM). The proposed study aims to bridge this research gap by identifying and ranking the barriers to GM implementation in this sector. Bayesian BWM offers a probabilistic framework that aggregates expert opinions with reduced bias, enhancing the robustness of barrier prioritization. The Bayesian probabilistic approach uses posterior distributions to draw reliable conclusions, even from limited data. This approach helps reduce uncertainty in expert-based decision-making, which is crucial for addressing complex sustainability challenges. Besides, no prior research has examined the challenges associated with the implementation of GM practices in the food and beverage industry from the perspective of an emerging economy. By structuring the barriers in clusters (organizational, financial, technological), the study also contributes to building a modular and scalable decision model that can be adapted across sectors. The findings support a growing body of work advocating for probabilistic, expert-based, and consensus-sensitive decision tools in sustainability science. Therefore, this work enhances both the methodological and theoretical repertoire for complex sustainability transitions in a data-sparse environment and emerging economy settings. The outcomes of this study can assist practitioners in setting strategies to overcome potential challenges related to GM implementation and promote sustainability in the industry. This study contributes to the food and beverage sector in the following ways: Theoretical contributions:

- Establishing a connection between GM and the food and beverage industry of an emerging economy like Bangladesh.
- Evaluating and ranking the identified challenges using Bayesian

BWM.

- Providing a theoretical basis for GM adoption to promote sustainability in the food and beverage sector.

Practical contributions:

- Identifying the noticeable challenges that impede most GM practices.
- Aiding the managers and policymakers to understand which barriers should be removed first for better performance.

The remaining sections are arranged in the following manner: Section 2 is comprised of a table of identified challenges and the methodology of Bayesian BWM. Section 3 presents the results and analysis. Section 4 discusses the findings and implications, while Section 5 concludes the study.

Methodology

Study context

Bangladesh's food and beverage industry has shown significant growth, contributing 22 % to the country's total manufacturing output and employing approximately 2.45 % of the labor force (Ashraf et al., 2017). Since 2000, this sector has emerged as a vital contributor to the national economy, exporting food and beverage products to 90 countries (Zim and Zahan, 2019). However, its competitive position is weakening due to increasing global and domestic competition (Ashraf et al., 2017). This indicates a need for strategic interventions to sustain growth in the face of evolving market dynamics.

The sector's environmental sustainability is increasingly under scrutiny, as decision-makers often overlook the environmental harm caused by manufacturing processes. Food production alone contributes 13.7 billion metric tons of carbon dioxide annually, accounting for 26 % of all anthropogenic greenhouse gas emissions (Poore and Nemecek, 2018). These emissions significantly impact biodiversity (Yang et al., 2024), reduce ecological resilience (Shi et al., 2022), and alter natural ecosystems (Wang and Azam, 2024). Addressing such issues requires adopting environmentally conscious manufacturing practices to mitigate the sector's ecological footprint.

Unfavorable interactions among food production systems, climate change, and ecological sustainability stem from the sector's high consumption of natural resources and energy (Sovacool et al., 2021). However, inadequate R&D funding and limited technical expertise hinder the development and implementation of innovative green manufacturing solutions (Nazir et al., 2019). For instance, effluent treatment processes in the beverage sector often rely on biological methods, which are underutilized despite their efficiency in managing organic waste (Kasmi, 2018). Enhancing R&D capacities and leveraging advanced technologies are critical to overcoming these limitations.

Small and medium-sized enterprises (SMEs) in this industry face unique challenges, such as limited access to sustainability resources and strategies (Takacs et al., 2022). Stable consumption patterns are juxtaposed with rising market competition, which compels SMEs to adopt circular models that track the entire lifecycle of products, from production to disposal (Dressler and Paunović, 2020). The absence of effluent treatment plants (ETPs) (Foglia et al., 2023), reliance on non-eco-friendly packaging materials (Dadas, et al., 2024), and emissions from manufacturing processes (Ehigiamusoe et al., 2024) exacerbate environmental damage. Stringent industrial regulations, if effectively implemented, could address these issues and ensure sustainable growth. Implementing green manufacturing practices can offer a

pathway to sustainable development in Bangladesh's food and beverage industry (Ahmad et al., 2024). By integrating environmental considerations into manufacturing, this sector can balance economic growth with ecological responsibility, ensuring long-term viability and global competitiveness.

Data collection and survey

This study applied Bayesian BWM to rank the identified challenges in green manufacturing implementation from an emerging economy perspective. At first, Challenges were identified using a systematic literature review and snowballing technique, focusing on Scopus, Web of Science, and Google Scholar databases. The search spanned studies published between 2022 and 2024. The research protocol, including databases, key terms, and inclusion/exclusion criteria, is outlined in Table 1.

Applying this protocol yielded 72 relevant studies, of which 34 research articles, projects, and reports were selected after a comprehensive full-text screening. From these sources, 18 potential challenges for GM adoption were initially identified in the food and beverage industry under the context of emerging economies. Then, a Google survey form was created to have experts' opinions on the primarily identified

Table 1
An overview of the systematic literature review protocol.

Research protocol	Detail description		department”—were excluded, while two additional challenges were incorporated: “High price of green raw materials and market competition” and “Difficulty in using green and renewable energy				
Databases	Google Scholar, Web of Science, Scopus						
Language	English						
Date range	2022–2024						
Determination of the challenges and clusters to implement GM	Total experts						
Search terms and keywords	Experience	Expertise area	Challenges of implementing GM in food” OR “GM	Designation	Expert number	Percentage	
X = 16	Above 12 years	Implementation, manufacturing economy” OR “GM and food industry barriers” OR “GM adoption challenges in food” OR “Limitations of GM in food” OR “factors hindering GM integration in food” OR “Barriers to incorporating GM in food manufacturing” AND “beverage”	Economics and consumer behavior	Professor	2	12.5	
				Chief Production Manager	1	6.25	
				Associate Director of R&D			
				Deputy Director of Sales			
				Professor			2
Exclusion criteria	(1) Irrelevant articles or articles lacking methodological and practical contributions, (2) Articles that emphasize quantitative methodologies to the exclusion of pertinent qualitative insights	Operations management	Chief Operations Manager	1	6.25		
			Associate Professor				
			Head of Production Planning				
			Research Scholar				
			Inclusion criteria	(A) Peer-reviewed articles relevant to the study RQs;(1) Include articles that explicitly address the RQs regarding challenges to GM adoption in the food and beverage industry; (2) Include studies that specifically investigate the GM;(B) Investigating Challenges and their effects: Include articles investigating and analyzing the challenges associated with implementing GM into the food and beverage manufacturing industry.	Production planning	Demand Forecasting Manager Assistant	2
Professor	1	6.25					
Senior Human Resource Advisor							
Senior Manager of Quality Assurance Assistant						2	12.5
Professor						1	6.25
Supply chain							
Ranking the challenges using the Bayesian Best Worst Method	6 to 8 years	Sustainability and environmental management	Research Scholar	1	6.25		
			Professor				
			Associate Director of R&D				
			Deputy Director of Sales				
			Y = 12	Above 12 years	Manufacturing	Professor	2
Chief Production Manager	1	8.33					
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method						6 to 8 years	Sustainability and environmental management
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning	Professor	1		
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning			Professor	1
Chief Production Manager							
Associate Director of R&D							
Deputy Director of Sales							
Ranking the challenges using the Bayesian Best Worst Method				6 to 8 years	Sustainability and environmental management	Professor	1
	Chief Production Manager						
	Associate Director of R&D						
	Deputy Director of Sales						
	Ranking the challenges using the Bayesian Best Worst Method	8 to 12 years	Demand planning				

sources". The final list of 17 challenges is detailed in Table B1 of Appendix B in the *supplementary materials*. These challenges were subsequently clustered into three major categories: organizational and governmental-related, financial and consumer behavior-related, and design and technology-related, as shown in Table 3.

In the second stage of expert feedback, the finalized challenges were ranked using Bayesian BWM. Experts were asked to construct "Best-to- Others" and "Others-to-Worst" matrices, rating the challenges on a scale of 1 to 9 (Shakur et al., 2024; Anam et al., 2025b). The initial 16 experts who participated in the validation phase were invited again for the Bayesian BWM evaluation. Among them, twelve experts responded (response

rate: 75 %). In the final stage, Bayesian BWM was applied to the experts' data to calculate the challenges' final weight and credal ranking. Then, the same 12 experts from stage two were approached again to provide valuable insights into the final results. Here, the response rate was 100 %. The framework of the study is presented in Fig. 2.

Bayesian Best-Worst Method (BWM)

The Bayesian BWM technique is employed in this study to evaluate the challenges for green manufacturing implementation in emerging

economies. Conventional BWM calculates a weight for each criterion

Table 3

List of challenges and clusters. using only one expert's preferences, but it cannot do so when dealing with several experts (Ahmed et al., 2023). The arithmetic mean approach can be employed to incorporate all expert evaluation scores, which could lead to mistakes in the weight findings. An improved variation of the initially introduced BWM is the Bayesian BWM (Munim et al., 2023). In Bayesian BWM, the opinions of several experts are combined within a probabilistic structure, allowing us to make judgments about the ultimately combined prioritization of criteria (Munim et al., 2022). Inconsistency is still a significant problem because of unstructured comparisons of factors. However, Bayesian BWM can overcome the problem by offering a higher consistency rate (Banihashemi et al., 2023). The following steps are required for Bayesian BWM (Debnath et al., 2023):

Cluster	Code	Challenges
Organizational and governmental-related challenges	O1	Lack of effective participation of the research and development (R&D) team
	O2	Lack of commitment from top management
	O3	Lack of adequate government policies and regulations
	O4	Inconsistency with the current infrastructure
	O5	Lack of external partnership and cooperation
	O6	Lack of availability of skilled workers
Financial and consumer behavior-related challenges	F1	High initial investment cost and lack of financial support
	F2	Absence of pressure from consumers
	F3	High price of green raw materials and market competition
	F4	Slow return on investment in green innovation
	F5	Lack of sense of corporate social responsibility
	F6	Lack of green awareness among customers and other stakeholders
Design and technology-related challenges	D1	Absence of a proper waste disposal system for the non-recyclable wastes
	D2	Complex design of green manufacturing system
	D3	Difficulty to incorporate eco-friendly design concepts
	D4	Absence of efficient recycling facilities
	D5	Difficulty to use green and renewable energy sources

The following steps are required for Bayesian BWM (Debnath et al., 2023):

- Best and Worst Criteria Selection:** Let, u be the number of experts evaluate a group of challenges $S = S_1, S_2, \dots, S_u$. At first, each expert selects the best challenge S^u_b and worst challenge S^u_w among all the challenges in the same cluster. A pairwise comparison is performed in the next step. Selected best and worst criteria can differ for each expert, where the best criteria present the most significance and the worst one presents the least significance
- Best to others pairwise comparison:** To construct a pairwise comparison matrix, each expert compares the best challenge S^u_b from the set of S challenge. Experts provide each challenge a number between 1 and 9 by comparing it with the best most influential challenge. The best challenge always score '1' when compared with itself and the worst criteria get a score of '9'. A score of '1' shows the challenge is equally importance,

and a '9' indicates S^u_b is more importance than the other challenge. Thus, "The Best-to-Others" vector A^u_b of the set of criteria from u number of experts

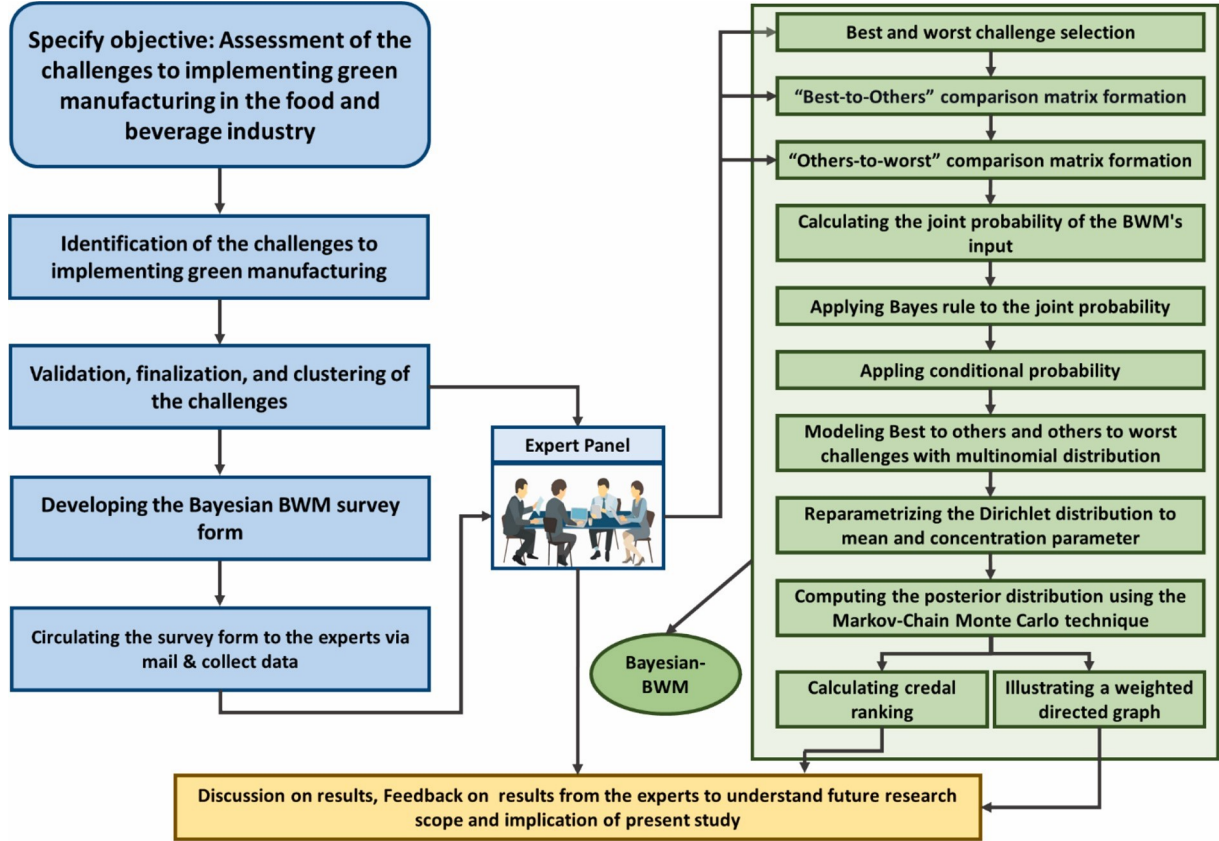


Fig. 2. Proposed research framework.

$$A_{B}^u = a_{B1}^u, a_{B2}^u, \dots, a_{Bu}^u, u = 1, 2, 3, \dots, U$$

Here, x and y represent arbitrary random variables.

For each $s_k \in S$, A_{Bk}^u represents the preference of the best criteria (S_B^u) over others by each expert. The “Best-to-Others” matrix of the main cluster and each of the clusters for each expert in this study are given in Tables C1-C4, Appendix C of the supplementary material.

3. Others to worst pairwise comparison: Similarly, experts compare pair-wisely other criteria in the set with the worst challenge S_w^u in a similar way to the one described in the previous step. When an expert assigns the value ‘9’ to a challenge, it indicates that the challenge is worse than the selected worst challenge S_w^u . The “Others-to-Worst” vector A_w^u of the set of criteria from u number of experts

$$A_{wW} = (a_{w1W}, a_{w2W}, \dots, a_{wuW})^T$$

For each $s_k \in S$, A_{kw}^u represents the preference of other challenges over the worst challenge (S_w^u) by each expert. The “Others-to-Worst” matrix of the main cluster and each of the clusters for each expert in this study is given in Tables D1-D4, Appendix D of the supplementary material.

4. Aggregated weight and individual optimal weight calculation: The aggregated optimal weight w_{agg} and each optimal weight $w_{1:U}$ for given $A_{1B:U}$ and $A_{1W:U}$ are calculated, where superscript 1:U indicates the total of all vectors in the base. Equations (3) and (4) represent the individual probability and joint probability distribution for each challenge, respectively

$$P(x) = \sum_y P(x, y)$$

$$P(w_{agg}, w_{1:U} | A_{1B:U}, A_{1W:U})$$

5. Bayesian hierarchical model: Considering all independence among different variables and applying the Bayes rule to the joint probability in equation (4), the results are combined in equation (5)

$$P(w_{agg}, w_{1:U} | A_{1B:U}, A_{1W:U}) \propto P(A_{1B:U}, A_{1W:U} | w_{agg}, w_{1:U}) P(w_{agg}, w_{1:U})$$

$$= P(w_{agg}) \prod_{u=1}^U P(A_{uw} | w_u) P(A_{ub} | w_u) P(w_u | w_{agg})$$

The probability chain rule, conditional independence of distinct variables, and the expert's individual assessments for each challenge are applied to derive the final equality. Before calculating the posterior distribution, all variables must be clearly defined. Since all variables in this context are integers, they can be modeled using a multinomial distribution. Furthermore, the two matrices (A_B^u and A_w^u) are inversely related, and the reverse weight is subsequently determined (equation 6 and 7).

$$A_B^u | w^u \text{ multinomial}(1/w^u), \forall u = 1, 2, \dots, U, (6) A_w^u | w^u \text{ multinomial}(w^u), \forall u = 1, 2, \dots, U, (7)$$

Then, aggregated weights of all u experts $w_{agg} = (w_1^*, w_2^*, \dots, w_v^*)$ and weight for each of the experts w^u are calculated by the following equations:

$$w^u | w^* \text{ Dir}(\gamma \times w^*), \forall u = 1, 2, \dots, U, (8) \gamma \text{ gamma}(0.1, 0.1)$$

$$(9) w^* \text{ Dir}(1) \quad (10)$$

Here, Dir exhibits the Dirichlet distribution, and gamma exhibits the gamma distribution where the shape parameter value is 0.1. As the probabilistic model defined by equation (10) does not provide a closed-form solution, a Markov-chain Monte

Carlo (MCMC) sampling can estimate the solution (Gallagher et al., 2009). w_k is the weight for each challenge (s_k), where s_k can be a random or stochastic event according to the probability concept and w_k shows the probability of the event occurring. In MCDM techniques, the weight vector $\sum_{k=1}^n w_k = 1$, and $w_k \geq 0$. $w_k = w_1'w_2'w_3', \dots$

$\dots, w_n, \dots, k-1$

To provide X Samples from probability distributions of the aggregated weight (w'), the Bayesian BWM is applied before that. Aggregated weight related to a challenge informs us of the importance and preference of that specific challenge over others. The formation and definition of credal ordering and credal ranking are based on a probabilistic perspective of aggregated weight (w').

Definition 1: Credal ordering (O) of two challenges, s_k and s_i , can be presented as follows:

$$O = (s_k, s_i, R, d)$$

Here, R represents the relationship, such as $>$, $<$, or $=$ between the performance measures s_k and s_i . $d \in [0,1]$ is the degree of confidence for the relations.

Definition 2: Credal ranking contains credal orderings of all pairs (s_k, s_i) for all $s_k, s_i \in S$ when a set of challenge ($S = S_1, S_2, \dots, S_n$) is given. Next, for every set of challenge s_k and s_i , the credal ordering degree calculation is required. To achieve this, the X samples were utilized with the online available tool JAGS and calculated the degree in the manner specified below:

$$P(s_k > s_i) = \frac{1}{X} \sum_{x=1}^X I(W_k^{(x)} > W_i^{(x)})$$

The Bayesian technique provides a confidence score for each credal ranking. Confidence score has a significant impact on experts' decisions as it provides them with more information. A Bayesian test predicated on the posterior distribution of aggregated weight finds a confidence score. When a challenge s_k is more important than another challenge s_i , the condition of $P(s_k > s_i) > 0.5$ is satisfied. The outcome of comparison among all the challenges can be presented in a weight-directed graph.

Results

This section presents the prioritized challenges for green manufacturing (GM) implementation as determined through the Bayesian BWM framework. The credal ranking of challenges is visualized using weighted directed graphs. Circular vertices (orange) represent the challenges, with their average weights

computed from the aggregated weight distributions (w'). Directed lines $S \rightarrow T$ indicates that challenge S is more significant than T , with a confidence score of q . These graphs depict the hierarchical structure of challenges based on their credal rankings.

Cluster-level rankings

The challenges were grouped into three main clusters: organizational and governmental, financial and consumer behavior, and design and technology. The credal ranking of these clusters is shown in Fig. 3. Among them, financial and consumer behavior challenges were the most critical, with an average weight of 0.3707. This was followed by organizational and governmental challenges (0.3218) and design and technology challenges (0.3075). While planning the strategies to overcome these challenges, top management should have a look at this cluster with the most importance. The design and technology and organizational and governmental clusters are almost close, though organizational and governmental get priority over design and technology. The financial and consumer behavior cluster demonstrated the highest confidence score (0.71) compared to the design and technology cluster, and a score of 0.69 compared to the organizational and governmental cluster. Conversely, the confidence score between organizational and governmental and financial and consumer behavior clusters (0.53) fell below the threshold value of 0.6, indicating variability in expert opinion (Ahmed et al., 2023). In this study, all values in the graph satisfy this threshold value except the confidence score between the organizational and governmental cluster and the financial and consumer behavior cluster. Related confidence scores for clusters based on probability are shown in Fig. 3.

Local rankings

Within the Financial and Consumer Behavior cluster (Fig. 4), the most significant challenge was identified as High initial investment cost and lack of financial support (0.1773). Other key challenges included the High price of green raw materials and market competition (0.1764), Slow return on investment in green innovation (0.1674), and lack of green awareness among customers and stakeholders (0.1617). Addressing these challenges requires infrastructure optimization and fostering consumer awareness, which is critical for promoting GM adoption. The graph exhibits the importance of each challenge compared to other challenges belonging to the financial and consumer cluster. Optimization between cost and infrastructure-equipment facilities can mitigate high initial investment costs. To deal with the last three consumer behavior-related challenges in this cluster, consumers' positive attitude toward green

C2: Financial and Consumer Behavior (0.3707)

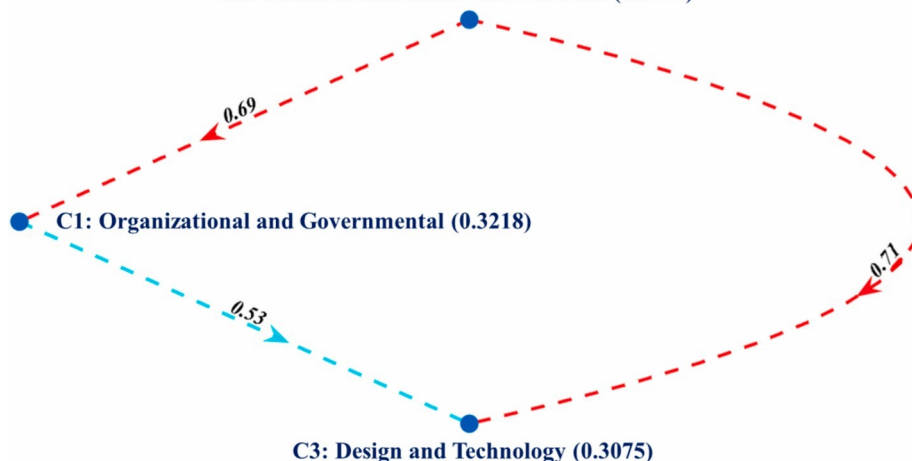


Fig. 3. Ranking of main clusters.

manufacturing is necessary. As food and beverage goods are related to health and manufacturing operations and have an impact on the environment, customer awareness will provide a competitive advantage to companies that are trying to implement green practices.

The Organizational and Governmental cluster (Fig. 5) ranked the lack of adequate government policies and regulations

F1. High initial investment cost and lack of financial support (0.1773)

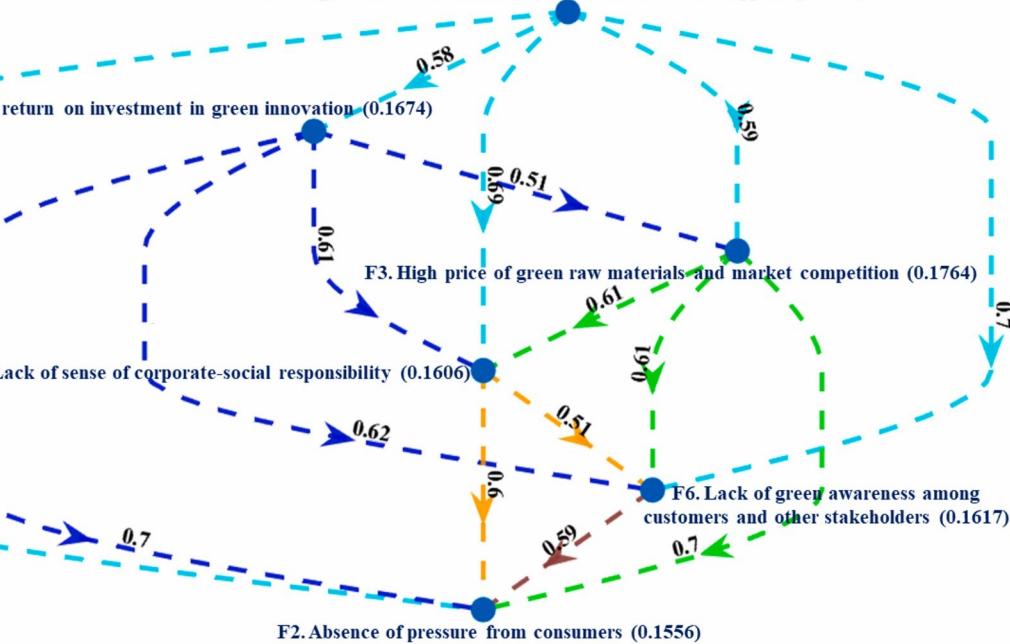


Fig. 4. Ranking of financial and consumer behavior-related challenges.

O3. Lack of adequate government policies and regulations (0.1984)

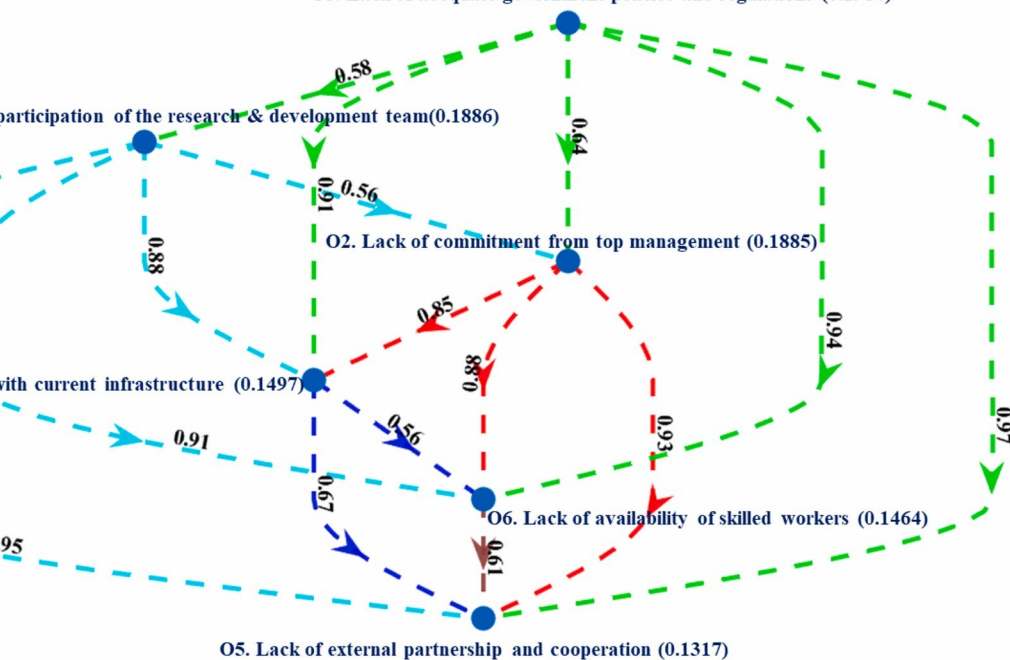


Fig. 5. Ranking of organizational and governmental-related challenges.

In the Design and Technology cluster (Fig. 6), the top-ranked challenge was Difficulty in incorporating eco-friendly design concepts (0.2405), followed by the Absence of efficient recycling facilities (0.2141), Complexity in designing GM systems (0.1972), and Absence of proper waste disposal systems (0.1932). Innovations in eco-friendly design and improved waste management systems are essential for addressing these obstacles.

(0.1984) as the top challenge, followed by the lack of effective participation from R&D teams (0.1886) and the lack of commitment from top management (0.1885). The inconsistency between infrastructure and GM requirements (0.1497) and the limited availability of skilled workers (0.1464) also emerged as significant challenges. Strengthening regulatory frameworks and encouraging industrial compliance can help mitigate these issues.

Global ranking

The confidence score metrics for the main clusters and all the sub-clusters are presented in Tables E1-E4 of Appendix E in the supplementary materials. The global ranking of the 17 challenges, calculated by combining cluster-level and local-level weights, is presented in Fig. 7. The top three global challenges were: "Difficulty in incorporating eco-friendly design concepts",

"Absence of efficient recycling facilities", and "High initial investment cost, and lack of financial support". These findings emphasize the multifaceted nature of challenges in GM adoption

D3. Difficulty to incorporate eco-friendly design concepts (0.2405)

and highlight priority areas for strategic interventions in the food and beverage industry.

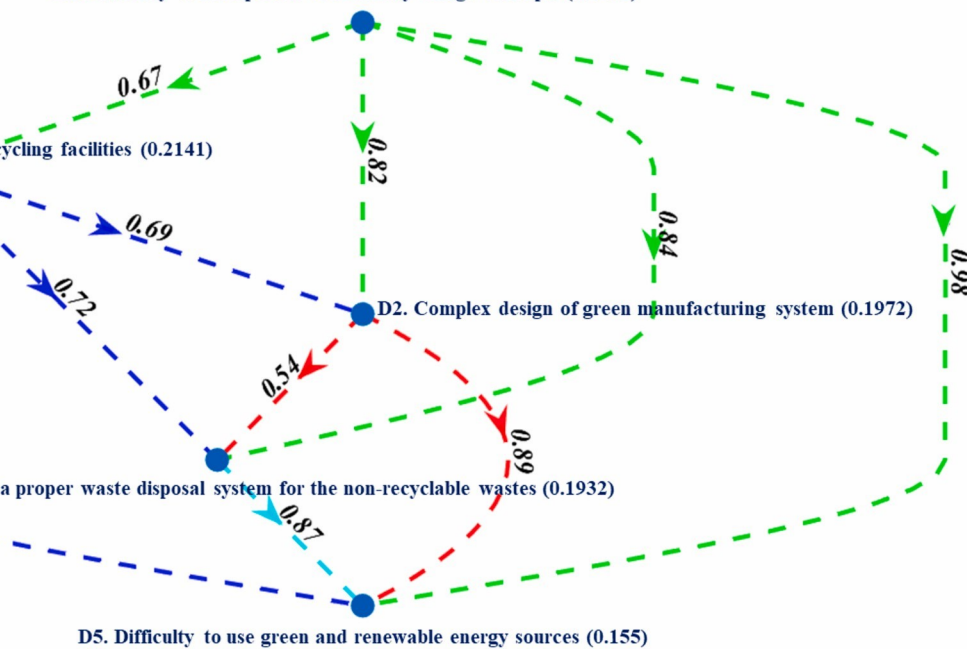


Fig. 6. Ranking of Design and Technology-related challenges.

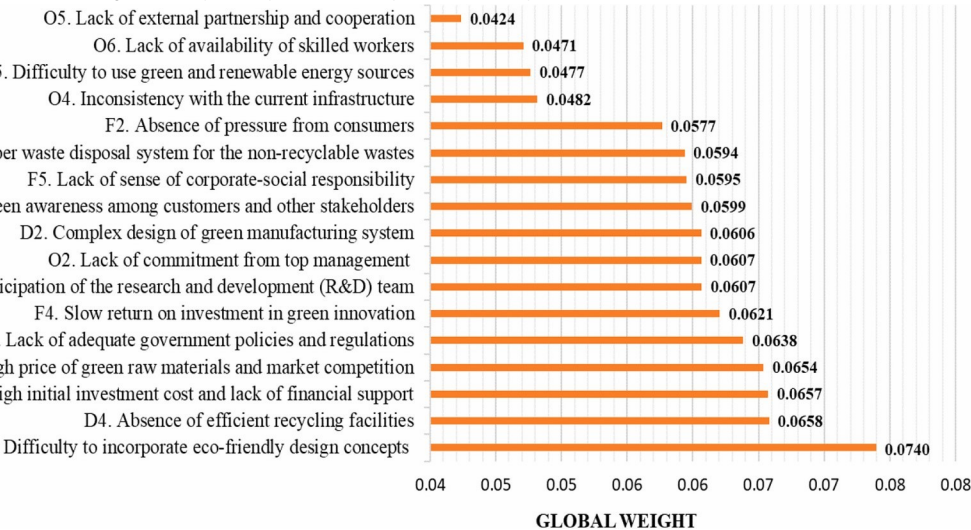


Fig. 7. Global ranking of identified challenges.

Discussions

The Bayesian Best Worst Method (BWM) identified "Difficulty to incorporate eco-friendly design concepts (D3)" as the most significant challenge to green manufacturing (GM) implementation in the food and beverage sector. This ranking reflects the technical and financial complexities associated with integrating eco-friendly practices into production systems in emerging economies (Li and Lin, 2024). Eco-friendly design concepts often require additional investments in research and development (R&D) (Batwara et al., 2022) and skilled personnel (Srisathan et al., 2023), which are scarce in these regions (Espinosa et al., 2021). According to recent studies on the Bangladeshi apparel manufacturing sector, the cost of setting up a green factory is approximately 25–30 % higher than that of a conventional factory (Mongabay, 2023; Imran et al., 2024). Moreover, there is also a need to develop educational and training programs for Bangladeshi workers to educate them on

the industry's impact on the environment and ecology (Al-Amin and Hoque, 2024). Hence, even though the initial cost of adopting green practices may deter many manufacturers (Khan et al., 2024), these measures provide long-term competitive advantages through cost savings (Abdullah and Lim, 2023), brand enhancement (Agu et al., 2024), and regulatory compliance (Agrawal et al., 2024).

To address this challenge, small factories could emulate sustainable practices from global leaders in eco-friendly manufacturing (Ahsan, 2024). Adopting initiatives like extensive Life Cycle Assessment (LCA), integrating green material index systems, and utilizing green design integration tools can help streamline the sustainability transition (Cerchione et al., 2024). Additionally, switching to sustainable packaging, such as recyclable plastics and compostable paper, can significantly reduce environmental impact (Asim et al., 2022). Innovative solutions like compostable paper trays, greener PET containers, and bio-based wrappers provide feasible options for the sector to

reduce its ecological footprint (Kong et al., 2022; Dybka-Stępien et al., 2021). For instance, Unilever Bangladesh is now using an innovative 'UREfill' system, which is a machine that is designed to reduce single-use plastic consumption (Unilever Bangladesh and Recognised, 2024).

The second most critical challenge, "Absence of efficient recycling facilities (D4)," reflects the inadequate infrastructure for recycling in emerging economies. Recycling systems are vital for reducing waste (Kurniawan et al., 2023), minimizing costs (Pimenov et al., 2022), and promoting sustainability in production (Hegab et al., 2023). However, the lack of proper waste treatment plants (Rosenboom et al., 2022), recycling systems for packaging (Awogbemi et al., 2022), and renewable energy use in factories exacerbate environmental pollution (Osman et al., 2023). In Bangladesh, inadequate budgets, infrastructure, and facilities for waste management have resulted in inefficient plastic disposal practices. Consequently, although 646 metric tons of plastic waste are collected daily, only 10 % is recycled (Abdullah and Abedin, 2024). Moreover, every year, approximately 1.11×10^7 metric tons of food waste are estimated to be produced in this country (Sarker et al., 2022). Implementing comprehensive recycling strategies, such as processing food industry waste in anaerobic digestion plants to produce compost and biogas, could mitigate these issues (Evide et al., 2021). Government incentives and recognition for effective recycling practices could encourage manufacturers to invest in such facilities (Taghipour et al., 2022). Furthermore, R&D teams should develop cost-efficient recycling methods tailored to local contexts to increase adoption rates (Laureti et al., 2024).

"High initial investment cost and lack of financial support (F1)" was ranked third, underscoring the significant financial burden faced by new factory owners. In developing economies, the cost of establishing green manufacturing systems, coupled with limited access to funding, poses a major obstacle (Kannan et al., 2022). Though the central bank of Bangladesh is promoting green financing, an underdeveloped bond market and limited venture capital make it difficult for Bangladeshi businesses to access funds for green manufacturing and sustainability initiatives (Azad et al., 2022). Policymakers can mitigate this issue by offering subsidies, tax incentives, and high-interest loans to businesses committed to sustainable practices (Gupta et al., 2022). Factory owners should focus on optimizing infrastructure, leveraging moderate-quality materials, and ensuring food safety to reduce initial expenditures (Ndzabukelwako et al., 2024). Additionally, emphasizing the long-term cost savings and environmental benefits of GM could attract investment from public and private stakeholders (Deshmukh and Tare, 2024).

The fourth-ranked challenge, "High price of green raw materials and market competition (F3)," stems from the higher costs associated with sustainable inputs and consumer preferences for cheaper products. While green raw materials increase production costs, they contribute to long-term environmental sustainability and can appeal to environmentally conscious consumers (Khan et al., 2020). Manufacturers can adopt green marketing strategies to highlight the benefits of eco-friendly products, which can attract customers willing to pay a premium (Savale et al., 2023). Promoting public awareness about the environmental impact of traditional manufacturing can also shift consumer behavior, driving demand for sustainable goods (Kar and Harichandan, 2022).

"Lack of adequate government policies and regulations (O3)" was identified as the fifth most significant barrier. While regulatory frameworks like ISO 14000 Environmental Management System (EMS) can encourage sustainable practices, weak enforcement limits their effectiveness (Ojo et al., 2024). Strict monitoring and auditing by third-party organizations can

ensure compliance with green manufacturing standards (Petrescu et al., 2021). Furthermore, capacity-building programs for policymakers and industry leaders can enhance the implementation and enforcement of green regulations.

The challenges ranked 6th, 7th, and 8th globally—Slow return on investment in green innovation (F4), lack of effective participation of the research and development (R&D) team (O1), and lack of commitment from top management (O2)—highlight critical obstacles to adopting green manufacturing (GM) in the food and beverage industry in emerging economies. In the context of emerging economies, where the "green concept" remains underdeveloped, a slow return on investment in green innovation (F4) acts as a deterrent for many industries (Aparna, 2021). To address this, implementing sustainability-based ROI models that track increased sales (Saleem et al., 2024), risk reduction (Gleißner et al., 2022), and cost savings is essential (Sabourin, 2021). Factory owners and policymakers must focus on promoting green products through awareness campaigns and collaborations with influential stakeholders to improve market acceptance and sales growth (Baah et al., 2021).

The lack of effective participation of R&D teams (O1) underscores the need for organizational support. Providing proper training facilities, modern laboratories equipped with advanced tools (Singh et al., 2023), and recognition of innovative contributions can empower R&D teams to develop eco-friendly design concepts and effective waste management systems (Ramadhan et al., 2024).

Similarly, lack of commitment from top management (O2) reflects an insufficient emphasis on sustainability initiatives in emerging economies. Offering incentives to top management and stakeholders can motivate them to adopt environmentally sustainable practices (Shahzad et al., 2023). Establishing partnerships with governmental organizations and leveraging policy frameworks to access resources can facilitate a stronger commitment to GM adoption (Ao et al., 2023).

Additionally, the Complex design of green manufacturing systems (D2) and the lack of green awareness among customers and other stakeholders (F6), ranked 9th and 10th, further emphasize the multifaceted nature of these challenges. The complex design of GM systems (D2) necessitates user-friendly technological solutions tailored to the specific needs of the food and beverage industry. Simplifying manufacturing systems through modular designs and accessible technologies can lower implementation barriers (Mykoniatis and Harris, 2021). Finally, the lack of green awareness among customers and stakeholders (F6) highlights the critical role of education and advocacy. Promoting the health and environmental benefits of green food and beverage products through marketing campaigns (Sembiring, 2021), public service announcements (Matos et al., 2023), and collaborations with non-governmental organizations can foster consumer demand (Prasanna et al., 2024), creating a market-driven push for GM adoption (Kadam, 2024).

Studies in other industries have highlighted similar barriers to GM implementation. For example, Virmani et al. (2021) ranked organizational barriers as the second most significant in the automobile manufacturing sector, while Sindhwani et al. (2019) identified lack of government funding as a major hurdle in adopting green agile production systems. The findings of Kalpande and Toke (2021) are similar to this work. According to them, "eco-design" is one of the indicators to enumerate enterprise green performance among the top three practiced indicators. This study prioritized "Difficulty to incorporate eco-friendly design concepts," "Absence of efficient recycling facilities," and "High initial investment cost and lack of financial support" as the top challenges in implementing GM in the food and beverage sector. The findings of this study are consistent with

those reported in other emerging economy contexts, reinforcing the broader applicability of the identified challenges. For instance, Kalpande and Toke (2021) found that eco- design was among the top three indicators used to assess enterprise green performance, aligning directly with our study's top-ranked challenge, "Difficulty to incorporate eco-friendly design concepts". Similarly, Li and Sarkis (2022) highlighted that China's slow adoption of eco- design practices has impeded the advancement of its green supply chains, a challenge likely to resonate in other developing nations as well.

In the Indian context, Karuppiyah et al. (2020) identified a lack of eco- design integration and financial constraints as key barriers to green manufacturing in SMEs, echoing our findings of "High initial investment cost" and "Absence of efficient recycling facilities". Likewise, Salim et al. (2018) reported that the Malaysian food and beverage sector avoid difficulties adopting ISO 14001 environmental management systems due to cost and regulatory burdens, consistent with our finding regarding the "Lack of adequate government policies and regulations". Additional studies further support the universality of these challenges. In South Africa, Du Plooy et al. (2022) identified financial limitations and infrastructural deficits as primary barriers to green manufacturing adoption among SMEs. In Vietnam, Huang et al. (2022) emphasized that government inaction and limited consumer awareness hinder green innovation, corresponding to our financial and consumer behavior cluster. Similarly, in Turkey, Kurtal et al. (2020) found that market competition and weak infrastructure limited sustainability practices in local food systems.

Also worth noting is that Agarwal et al. (2024), who examined the Indian automotive sector, also found financial constraints to be the most pressing challenge for green smart manufacturing. These parallels affirm that the difficulties faced in the Bangladeshi food and beverage sector are not isolated; they reflect a broader trend across emerging economies where financial, technological, and policy-related barriers impede the transition to sustainable production practices. By addressing these challenges through targeted strategies and enabling mechanisms, emerging economies can accelerate their transition to greener, more resilient manufacturing systems.

Theoretical implications

This study provides vital theoretical contributions by addressing the challenges associated with GM adoption in the food and beverage sector, particularly within emerging economies. Unlike prior studies that focus on critical success factors, this research prioritizes and ranks GM barriers using the Bayesian Best Worst Method (BWM) framework, offering a structured and hierarchical understanding of these obstacles. Unlike traditional decision-making methods, the Bayesian BWM incorporates probability distributions to effectively model uncertainty in expert judgments. This approach allows the experts to express their confidence levels in their preferences and offers a stronger and more reliable basis for group decision-making, which in turn enhances the robustness of the analysis. Thus, by accommodating uncertainty and variability in human judgment, the Bayesian framework offers a more realistic and probabilistic representation of potential outcomes (Mohammadi and Rezaei, 2020; Modares et al., 2025).

The study identifies "difficulty to incorporate eco-friendly design concepts" and the "absence of efficient recycling facilities" as the two most critical challenges. These findings align with previous works, such as Kalpande and Toke (2021), emphasizing the growing significance of eco-friendly practices like green product design, sustainable packaging, and recycling processes.

This study advances theoretical discussions by highlighting tools like Life Cycle Assessment (LCA) and green material index systems, demonstrating their potential to enhance operational efficiency and reduce environmental impact.

Focusing on emerging economies, this study sheds light on challenges such as high initial investment costs, limited financial support, and weak policy frameworks (Anand et al., 2021). These barriers are particularly critical in resource-constrained settings, where factory owners often struggle to balance profitability with sustainability initiatives, as noted by Adams et al. (2023). Moreover, the reluctance of governing bodies to enforce stringent regulations further impedes GM adoption, emphasizing the need for institutional support, consistent with Ullah et al. (2022).

This research contributes to the theoretical understanding of green innovation by addressing the interplay between financial constraints (Zhang and Vigne, 2021), market dynamics (Yuan and Cao, 2022), and consumer behavior (Oduro et al., 2022). Challenges like the high cost of green raw materials and slow returns on green investments highlight the importance of sustainability-based Return on Investment (ROI) models and green marketing strategies (Braik et al., 2024). These approaches can help firms balance short-term market pressures with long-term environmental and economic benefits (Xu et al., 2023).

Finally, this study's focus on the food and beverage sector in an emerging economy offers a unique perspective, bridging a critical gap in the literature. By providing a ranked list of challenges and actionable insights, the research offers a foundation for future studies to explore advanced technologies, cross-industry collaborations, and policy interventions to overcome these barriers (Luthra et al., 2022). This contribution enriches the theoretical discourse on GM by emphasizing its multidimensional nature and paving the way for sustainable manufacturing practices.

Practical implications

Effective strategies devised by top management play a pivotal role in driving industrial progress while ensuring environmental sustainability. In emerging economies, however, limited commitment from top management to social and environmental responsibilities often disrupts ecological balance (Attah-Boakye et al., 2022). This study highlights that consumer-driven pressure for green products can hold policymakers and industry leaders more accountable for implementing sustainable practices (Halonen, 2024).

Managers should prioritize the efficient disposal of non-recyclable wastes, the adoption of advanced recycling facilities, and sustainable infrastructure improvements. Incorporating tools like Life Cycle Assessment (LCA), green material indices, and sustainable packaging designs can align production processes with eco-friendly objectives (Jagoda et al., 2023). Strategies such as using recyclable plastics, compostable packaging, and adopting minimal-waste manufacturing can reduce environmental impact while enhancing business competitiveness (Jain et al., 2024). An illustrative example can be drawn from Nestlé, which has employed LCA to analyze the environmental impacts of its food and beverage products and has transitioned to recyclable and compostable packaging in several consumer market segments (Kandasamy et al., 2025). Similarly, Hindustan Unilever Limited's "Sustainable Living Plan" has demonstrated how aligning sustainability goals with the company's core business strategy and operations can reduce greenhouse gas emissions, water usage, and waste across its operations, and strengthen its brand reputation, enhance consumer loyalty, and attract investment from socially conscious investors (P Shravan and Vaishali, 2024). A study by Gazzola et al. (2024) highlighted how large European

food and beverage companies like Danone, Heineken, and Diageo have embedded sustainability into their core business strategies through the adoption of renewable energy, water efficiency initiatives, and sustainable packaging practices, reinforcing the idea that sustainable production practices can drive long-term value and competitive advantage. In Bangladesh, companies such as Pran Foods Ltd., Akij Food & Beverage Ltd., and Aarong Dairy have also taken steps toward green manufacturing by maintaining effluent treatment plants at production sites, exploring biodegradable packaging options, promoting waste minimization, and adopting clean energy options (Haq et al., 2021; Mollah et al., 2024).

Moreover, adhering to government policies and regulations, such as Environmental Management Systems (e.g., ISO 14001), can attract financial support and foster public trust (Ojiako et al., 2024). These compliance measures can also encourage investment by showcasing the industry's commitment to sustainability (Ayorinde et al., 2024). To address high initial investment costs, managers should seek partnerships with governmental and private stakeholders to secure more funding and technical support. Recently, the Government of Bangladesh, with support from the World Bank, has developed a comprehensive green growth strategy (World Bank, 2024). The government collaborated with development partners and implemented the Third Country Investment Plan (CIP3) for 2021–2025, which mobilizes resources towards sustainable, nutrition-sensitive, and resilient investments in the food system (Food and Agriculture Organization, 2025). Moreover, an initiative of the World Economic Forum, the Global Plastic Action Partnership (GPAP), collaborates with Bangladesh to develop and implement the National Action Plan for Sustainable Plastic Management. This plan sets ambitious goals, including recycling 50 % of plastics by 2025, reducing single-use plastics by 90 % by 2026, and cutting plastic waste generation by 30 % by 2030 (World Bank, 2021). From October 2024, the Bangladeshi government enforced a ban on the use of polythene and polypropylene bags (Environment and Social Development Organization, 2024). Bangladesh Bank is currently promoting investments in green projects through several refinancing schemes and funds, such as the Green Transformation Fund (GTF) and the Technology Development/Upgradation Fund (TDF). In its national budget, Bangladesh also offered a 2-percentage-point tax reduction as an incentive for green factories (Bangladesh Foreign Trade Institute, 2024). All of these reflect how Bangladesh is making concerted efforts to promote sustainable manufacturing practices.

Ultimately, the strategic focus should aim to save time, energy, and costs while optimizing processes across the production lifecycle (Dahmani et al., 2021). Promoting the benefits of green food and beverage products through targeted marketing strategies can further drive consumer demand and competitive advantage (Adams et al., 2023). By integrating these strategies, top management can ensure sustainable growth and long-term environmental and economic viability for the food and beverage sector.

Implications for achieving SDGs

Green Manufacturing (GM) addresses the economic, social, and environmental impacts of traditional manufacturing practices, aligning with the goals of sustainability. The findings of this study offer valuable insights for policymakers and governing bodies in designing operational strategies that contribute to achieving Sustainable Development Goals (SDGs). This study particularly emphasizes SDG 9 (Industry, Innovation, and Infrastructure), SDG 12 (Responsible Consumption and

Production), SDG 7 (Affordable and Clean Energy), and SDG 17 (Partnership for the Goals).

SDG 9 can be achieved through the modernization and transformation of industrial structures using GM practices. Integrating renewable energy into manufacturing processes, adopting eco-friendly designs, and implementing sustainable technologies promote environmentally friendly industrialization. These practices also foster innovation and advance infrastructure development, enabling the food and beverage industry to become more resilient and sustainable.

SDG 12 emphasizes responsible consumption and production, achievable through strategies such as efficient recycling, waste reduction, and the use of green raw materials. Educating stakeholders, including consumers, on the benefits of sustainable products and promoting corporate social responsibility can drive demand for green innovations. Additionally, adopting tools like Life Cycle Assessment (LCA) ensures the sustainability of the entire production process.

SDG 7 calls for affordable and clean energy, which can be addressed by evaluating and implementing renewable energy solutions like solar photovoltaics, wind energy, and bioenergy. This study highlights the role of R&D teams in identifying cost-effective renewable energy sources to enhance energy efficiency and maintain ecological balance.

SDG 17 focuses on partnerships and collaborations to overcome financial and technical barriers. Partnerships with global and local organizations, along with government financial support, can address high initial investment costs and promote shared expertise in sustainable practices. Encouraging cross-sector collaboration accelerates the adoption of green technologies and infrastructure improvements.

Conclusion

GM in the food and beverage industry significantly benefits businesses and the environment. However, in emerging economies, a lack of consumer pressure due to limited green awareness often results in manufacturers persisting with traditional methods to reduce costs and avoid adopting new technologies. This study highlights the need for policymakers and industry leaders to prioritize green practices as a strategic guideline for addressing sector-specific challenges. With the rising demand for consumer goods and beverages, it is imperative for emerging economies to embrace green innovations and eco-friendly design concepts to achieve environmental, economic, and social sustainability.

Using a Bayesian Best-Worst Method (BWM) framework, this study identified the “Financial and consumer behavior” cluster as the most critical area requiring attention. Strategies such as understanding customer demand patterns, enhancing customer loyalty, and promoting “green achievements” can drive consumer interest in sustainable practices. Additionally, the study revealed that “Difficulty to incorporate eco-friendly design concepts,” “Absence of efficient recycling facilities,” and “High initial investment cost and lack of financial support” are the top three challenges hindering GM adoption in this sector. Incorporating lean tools like Kaizen, Eight Waste Elimination, and the 5S methodology with eco-friendly designs can ensure energy efficiency, optimal resource utilization, and cost reduction across production stages.

Despite its contributions, this study has some limitations that can be overcome in future attempts. For instance, reliance on a small sample of GM-aware professionals and the potential for judgment biases is an important limitation of this study. A larger sample size and assigning weights to experts based on their experience could improve accuracy to a greater extent. Moreover,

the study has explored GM challenges, focusing on the food and beverage industries of Bangladesh. Future research should explore GM challenges across different economic systems, such as capital or mixed economies, and industries like automotive, steel, cement, textile and apparel, furniture, and biomedical sectors. Again, the study considers only 17 challenges, which may evolve or expand over time as green manufacturing practices and industry conditions change. Further studies can adopt longitudinal approaches to continuously reassess and refine the set of identified challenges. Since the study uses Bayesian BWM, the challenges are assumed to be independent, which may not hold in complex decision-making scenarios. Future studies could integrate Bayesian BWM with Bayesian Belief Networks or other machine learning frameworks to better capture and visualize interdependencies among challenges. Furthermore, advanced decision-making theories, such as Intuitionistic, Spherical Fuzzy, and Pythagorean Fuzzy theories, could reduce uncertainty in prioritization processes. Advanced methodologies, including m-TISM, PLS-SEM, and DEMATEL, could also enhance future evaluations. Addressing these limitations and broadening the research scope will foster a deeper understanding of GM practices and accelerate sustainable transformations across various manufacturing industries.

Funding acknowledgment

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Maishat Lubaba: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Sanjida Hassan:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Munera Sultana:** Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Shah Murtoza Morshed:** Writing – review & editing, Visualization, Validation, Methodology, Investigation. **A.B.M. Mainul Bari:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the experts who took their valuable time to provide feedback for this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wmb.2025.100225>.

References

- Abdullah, M., Abedin, M.Z., 2024. Assessment of plastic waste management in Bangladesh: a comprehensive perspective on sorting, production, separation, and recycling. *Results Surf. Interfaces* 15, 100221.
- Abdullah, N., Lim, A., 2023. The incorporating sustainable and Green IT practices in modern IT service operations for an environmentally conscious future. *J. Sustain. Technol. Infrastruct. Plan.* 7 (3), 17–47.
- Adam, D.H., Ende Supriyadi, Y.N., Siregar, Z.M.E., 2020. Green manufacturing, green chemistry and environmental sustainability: a review. *Int. J. Sci. Technol. Res.* 9, 2209–2211.
- Adams, D., Donovan, J., Topple, C., 2023. Sustainability in large food and beverage companies and their supply chains: an investigation into key drivers and barriers affecting sustainability strategies. *Bus. Strateg. Environ.* 32 (4), 1451–1463.
- Agarwal, S., Saxena, K.K., Agrawal, V., Dixit, J.K., Prakash, C., Buddhi, D., Mohammed, K.A., 2024. Prioritizing the barriers of green smart manufacturing using AHP in implementing Industry 4.0: a case from Indian automotive industry. *The TQM Journal* 36 (1), 71–89.
- Agrawal, R., Agrawal, S., Samadhiya, A., Kumar, A., Luthra, S., Jain, V., 2024. Adoption of green finance and green innovation for achieving circularity: an exploratory review and future directions. *Geosci. Front.* 15 (4), 101669.
- Agu, E.E., Iyelolu, T.V., Idemudia, C., Ijomah, T.I., 2024. Exploring the relationship between sustainable business practices and increased brand loyalty. *Int. J. Manage. Entrepreneurship Res.* 6 (8), 2463–2475.
- Ahamad, H., Alam, O., Sultana, N., Misbahuzzaman, K., 2024. Current environmental management practices in major manufacturing industries of Chattogram, Bangladesh. *Environ. Develop. Sustain.* 1–27.
- Ahmed, T., Karmaker, C.L., Nasir, S.B., Moktadir, M.A., 2023. Identifying and analysis of key flexible sustainable supply chain management strategies toward overcoming the post-COVID-19 impacts. *Int. J. Emerg. Mark.* 18 (6), 1472–1492.
- Ahsan, M.J., 2024. Green leadership and innovation: catalysts for environmental performance in Italian manufacturing. *Int. J. Organ. Anal.*
- Akanmu, M.D., Nordin, N., 2022. Hybrid lean practices integrated with IR 4.0 and sustainability in Malaysia food and beverages companies: conceptual framework and hypothesis development. *Pertanika J. Sci. Technol.* 30 (3).
- Akash, M.H., Al Aziz, R., Karmaker, C.L., Islam, A.R.M.T., 2025. Investigating the attributes for implementing circular economy in the textile manufacturing supply chain: implications for the triple bottom line of sustainability. *Sustainable Horiz.* 14, 100129.
- Alaghemandi, M., 2024. Sustainable solutions through innovative plastic waste recycling technologies. *Sustainability* 16 (23), 10401.
- Alam, M.J., Reza, S.A., Ogawa, K., Ahsan, A.H.M., 2024. Sustainable employment for vocational education and training graduates: the case of future skills matching in Bangladesh. *Int. J. Train. Res.* 1–23.
- Al-Amin, A.Q., Hoque, M.R., 2024. Bangladesh green industries diagnostic.
- Ali, S.S., Kaur, R., Khan, S., 2023. Evaluating sustainability initiatives in warehouse for measuring sustainability performance: an emerging economy perspective. *Ann. Operat. Res.* 324 (1), 461–500.
- Amrutha, V.N., Geetha, S.N., 2020. A systematic review on green human resource management: Implications for social sustainability. *J. Clean. Prod.* 247, 119131.
- An, H., Razaq, A., Nawaz, A., Noman, S.M., Khan, S.A.R., 2021. Nexus between green logistic operations and triple bottom line: evidence from infrastructure-led Chinese outward foreign direct investment in Belt and Road host countries. *Environ. Sci. Pollut. Res.* 28 (37), 51022–51045.
- Anam, M.Z., Islam, M.H., Islam, M.T., Bari, A.M., Raihan, A., 2025a. A Fermatean fuzzy approach to analyze the drivers of digital transformation in the agricultural production sector: a pathway to sustainability for emerging economies. *Green Technol. Sustain.* 3 (3), 100197.
- Anam, M.Z., Shakur, M.S., Bari, A.M., Debnath, B., Bristy, F.T., 2025b. Exploring the barriers to decarbonizing the transportation system: a pathway to a cleaner future in emerging economies. *Innovat. Green Dev.* 4 (4), 100256.
- Anand, J., McDermott, G., Mudambi, R., Narula, R., 2021. Innovation in and from emerging economies: New insights and lessons for international business research. *J. Int. Bus. Stud.* 52, 545–559.
- Ao, X.Y., Ong, T.S., Aprile, R., Di Vaio, A., 2023. Environmental uncertainty and digital technologies corporate in shaping corporate green behavior and tax avoidance. *Sci. Rep.* 13 (1), 22170.
- Aparna, V.R., 2021. Agricultural knowledge management initiatives by ICAR: an overview. *J. Agric. Extens. Manage.* 22 (2), 175.
- Ashraf, S.R.B., Rashid, M.M., Rashid, A.H., 2017. Implementation of 5S methodology in a food & beverage industry: a case study. *Int. Res. J. Eng. Technol.* 4 (3), 1791–1796.
- Asim, Z., Shamsi, I.R.A., Wahaj, M., Raza, A., Abul Hasan, S., Siddiqui, S.A., Seng Teck, T., 2022. Significance of sustainable packaging: a case-study from a supply chain perspective. *Appl. Syst. Innovation* 5 (6), 117.
- Atieh, A.M., Cooke, K.O., Osiyevskyy, O., 2023. The role of intelligent manufacturing systems in the implementation of Industry 4.0 by small and medium enterprises in developing countries. *Eng. Res.* 5 (3), e12578.
- Attah-Boakye, R., Adams, K., Yu, H., Koukpaki, A.S.F., 2022. Eco-environmental footprint and value chains of technology multinational enterprises operating in emerging economies. *Strateg. Chang.* 31 (1), 99–116.
- Awewomom, J., Dzeble, F., Takyl, Y.D., Ashie, W.B., Ettay, E.N.Y.O., Afua, P.E., Akoto, O., 2024. Addressing global environmental pollution using environmental control techniques: a focus on environmental policy and preventive environmental management. *Discover Environment* 2 (1), 8.

- Awogbemi, O., Kallon, D.V.V., Bello, K.A., 2022. Resource recycling with the aim of achieving zero-waste manufacturing. *Sustainability* 14 (8), 4503.
- Ayorinde, O.B., Daudu, C.D., Etukudoh, E.A., Adefemi, A., Adekoya, O.O., Okoli, C.E., 2024. Climate risk assessment in petroleum operations: a review of CSR practices for sustainable resilience in the United States and Africa. *Eng. Sci. Technol. J.* 5 (2), 385–401.
- Azad, M.A.K., Islam, M.A., Sobhani, F.A., Hassan, M.S., Masukujjaman, M., 2022. Revisiting the current status of green finance and sustainable finance disbursement: a policy insights. *Sustainability* 14 (14), 8911.
- Baah, C., Opoku-Agyeman, D., Acquah, I.S.K., Agyabeng-Mensah, Y., Afum, E., Faibil, D., Abdoulaye, F.A.M., 2021. Examining the correlations between stakeholder pressures, green production practices, firm reputation, environmental and financial performance: evidence from manufacturing SMEs. *Sustainable Prod. Consumption* 27, 100–114.
- Bangladesh Foreign Trade Institute. Trade Benefits of Green Industry Initiatives in Bangladesh. Available online: <https://www.bfti.org.bd/blog-details/15> (accessed on 15 November, 2024).
- Banihashemi, S.A., Khalilzadeh, M., Antucheviciene, J., Edalatpanah, S.A., 2023. Identifying and prioritizing the challenges and obstacles of the green supply chain management in the construction industry using the fuzzy BWM method. *Buildings* 13 (1), 38.
- Bapat, G.S., Kumar, A., Kumar, A., Hota, S.L., Kavita, Singh, K., 2024. Sustainable green manufacturing approaches in India—a step towards a new green revolution through SMEs. *Engineering Proceedings* 59 (1), 211.
- Batwara, A., Sharma, V., Makkar, M., Giallanza, A., 2022. An empirical investigation of green product design and development strategies for eco industries using kano model and fuzzy AHP. *Sustainability* 14 (14), 8735.
- Bendig, D., Kleine-Stegemann, L., Gisa, K., 2023. The green manufacturing framework—A systematic literature review. *Cleaner Eng. Technol.* 13, 100613.
- Braik, A., Saleh, Y., Jaaron, A.A., 2024. Green marketing practices and organizational sustainable performance in developing countries context: an empirical study. *J. Foodserv. Bus. Res.* 27 (6), 769–809.
- Cerchione, R., Morelli, M., Passaro, R., Quinto, I., 2024. A critical analysis of the integration of life cycle methods and quantitative methods for sustainability assessment. *Corp. Soc. Respon. Environ. Manag.*
- Chen, X., Voigt, T., 2020. Implementation of the Manufacturing Execution System in the food and beverage industry. *J. Food Eng.* 278, 109932.
- Cheng, H., Chaudhry, H.R., Kazi, I., Umar, M., 2024. Unlocking greener supply chains: a global innovative perspective on the role of logistics performance in reducing ecological footprints. *J. Innov. Knowl.* 9 (4), 100612.
- Chien, F., Ngo, Q.T., Hsu, C.C., Chau, K.Y., Iram, R., 2021. Assessing the mechanism of barriers towards green finance and public spending in small and medium enterprises from developed countries. *Environ. Sci. Pollut. Res.* 28 (43), 60495–60510.
- Chowdhury, M.T., Sarkar, A., Paul, S.K., Moktadir, M.A., 2022. A case study on strategies to deal with the impacts of COVID-19 pandemic in the food and beverage industry. *Oper. Manag. Res.* 15 (1), 166–178.
- Dadas, A.B., Demircan, B., Velioglu, Y.S., 2024. Recent advances in water-soluble edible pouches: production methods, industrial applications, and legal regulations. *Akademik Gıda* 22 (2), 143–157.
- Dahmani, N., Benhida, K., Belhadi, A., Kamle, S., Elfezazi, S., Jauhar, S.K., 2021. Smart circular product design strategies towards eco-effective production systems: a lean eco-design industry 4.0 framework. *J. Clean. Prod.* 320, 128847.
- D'Angelo, V., Cappa, F., Peruffo, E., 2023. Green manufacturing for sustainable development: the positive effects of green activities, green investments, and non-green products on economic performance. *Bus. Strateg. Environ.* 32 (4), 1900–1913.
- Das, S., Singh, C.K., Sodhi, K.K., Singh, V.K., 2023. Circular economy approaches for water reuse and emerging contaminant mitigation: innovations in water treatment. *Environ., Develop. Sustain.* 1–42.
- Debnath, B., Shakur, M.S., Bari, A.M., Karmaker, C.L., 2023. A Bayesian Best–Worst approach for assessing the critical success factors in sustainable lean manufacturing. *Decision Anal. J.* 6, 100157.
- Deshmukh, P., Tare, H., 2024. Green marketing and corporate social responsibility: a review of business practices. *Multidisciplinary Reviews* 7 (3), 2024059.
- Dressler, M., Paunović, I., 2020. Towards a conceptual framework for sustainable business models in the food and beverage industry: the case of German wineries. *Br. Food J.* 122 (5), 1421–1435.
- Du Plooy, S., Neethling, K., Nel, A., Nel, J.D., 2022. Drivers of and barriers to green manufacturing in South Africa. *J. Contemporary Manage.* 19 (1), 260–298.
- Dybka-Stępien, K., Antolak, H., Kmietek, M., Piechota, D., Koziński, A., 2021. Disposable food packaging and serving materials—Trends and biodegradability. *Polymers* 13 (20), 3606.
- Ehigiamusoe, K.U., Lean, H.H., Somasundram, S., 2024. Analysis of the environmental impacts of the agricultural, industrial, and financial sectors in Malaysia. *Energy Environ.* 35 (5), 2329–2356.
- Environment and Social Development Organization. Bangladesh Set to Enforce Ban on Polythene Bags in Supermarkets from October 1st. Available online: <https://bansup.esdo.org/bangladesh-set-to-enforce-ban-on-polythene-bags-in-supermarkets-from-october-1st/> (accessed on 10 September, 2024).
- Escobar, A., Luna, J., Caraballo, A., 2023. Barriers to sustainable green innovation in meeting the challenges of the global economy of firms. *Global J. Environ. Sci. Manage.* 9, 219–232.
- Espinosa, R.V., Soto, M., Garcia, M.V., Naranjo, J.E., 2021. Challenges of implementing cleaner production strategies in the food and beverage industry: literature review. *Advances and Applications in Computer Science, Electronics and Industrial Engineering: Proceedings of CSEI 2020*, 121–133.
- Evode, N., Qamar, S.A., Bilal, M., Barcelo, D., Iqbal, H.M., 2021. Plastic waste and its management strategies for environmental sustainability. *Case Stud. Chem. Environ. Eng.* 4, 100142.
- Fahad, S., Alnori, F., Su, F., Deng, J., 2022. Adoption of green innovation practices in SMEs sector: evidence from an emerging economy. *Econom. Res.-Ekonomska Istraživanja* 35 (1), 5486–5501.
- Feng, Y., Wang, X., Liang, Z., 2021. How does environmental information disclosure affect economic development and haze pollution in Chinese cities? The mediating role of green technology innovation. *Sci. Total Environ.* 775, 145811.
- Foglia, A., Gonzalez-Camejo, J., Radini, S., Sgroi, M., Li, K., Eusebi, A.L., Fatone, F., 2023. Transforming wastewater treatment plants into reclaimed water facilities in water-unbalanced regions: an overview of possibilities and recommendations focusing on the Italian case. *J. Clean. Prod.* 410, 137264.
- Food and Agriculture Organization. Bangladesh Third Country Investment Plan Sustainable, Nutrition-Sensitive and Resilient Food Systems (2021–2025). Available online: <https://www.fao.org/faolex/results/details/es/c/LEX-FAOC228853> (accessed on 24 January, 2025).
- Gallagher, K., Karvin, K., Nielsen, S., Sambridge, M., Stephenson, J., 2009. Markov chain Monte Carlo (MCMC) sampling methods to determine optimal models, model resolution and model choice for Earth Science problems. *Mar. Pet. Geol.* 26 (4), 525–535.
- Gazzola, P., Pavione, E., Amelio, S., Mauri, M., 2024. Sustainable strategies and value creation in the food and beverage sector: the case of large listed European companies. *Sustainability* 16 (22), 9798.
- Gleißner, W., Günther, T., Walkshausl, C., 2022. Financial sustainability: measurement and empirical evidence. *J. Bus. Econ.* 92 (3), 467–516.
- Guan, L., Li, W., Guo, C., Huang, J., 2023. Environmental strategy for sustainable development: Role of digital transformation in China's natural resource exploitation. *Resour. Policy* 87, 104304.
- Gull, M., Rashid, M., Hassan, S., Rehman, S., 2024. Role of top management green commitment, adaptability culture and green product innovation in boosting organizational green performance in Pakistan. *Res. J. Text. Appar.* 28 (4), 1066–1090.
- Guo, H., Dong, M., Tsinopoulos, C., Xu, M., 2024. The influential capacity of carbon neutrality environmental orientation in modulating stakeholder engagement toward green manufacturing. *Corp. Soc. Respon. Environ. Manag.* 31 (1), 292–310.
- Gupta, A., Alston, L., Needham, C., Robinson, E., Marshall, J., Boelsen-Robinson, T., Peeters, A., 2022. Factors influencing implementation, sustainability and scalability of healthy food retail interventions: a systematic review of reviews. *Nutrients* 14 (2), 294.
- Haleem, A., Javaid, M., Singh, R.P., Suman, R., Qadri, M.A., 2023. A pervasive study on Green Manufacturing towards attaining sustainability. *Green Technol. Sustainability* 1 (2), 100018.
- Halonen, H., 2024. Why do companies go green? Examining the drivers of sustainable self-regulation.
- Haq, M.K., Arman, S.M., Nazera, F., 2021. Private enterprise and the sustainable community economic development in Bangladesh a case study on the PRAN-RFL Group. *Int. J. Res. Innovation Soc. Sci.* 5 (4), 49–55.
- Hariram, N.P., Mekha, K.B., Suganthan, V., Sudhakar, K., 2023. Sustainability: an integrated socio-economic-environmental model to address sustainable development and sustainability. *Sustainability* 15 (13), 10682.
- Hegab, H., Shaban, I., Jamil, M., Khanna, N., 2023. Toward sustainable future: strategies, indicators, and challenges for implementing sustainable production systems. *Sustain. Mater. Technol.* 36, e00617.
- Hossain, M., Al Aziz, R., Karmaker, C.L., Debnath, B., Bari, A.M., Islam, A.R.M.T., 2024. Exploring the barriers to implement industrial symbiosis in the apparel manufacturing industry: implications for sustainable development. *Heliyon* 10 (13).
- Huang, Y.F., Chen, A.P.S., Do, M.H., Chung, J.C., 2022. Assessing the barriers of green innovation implementation: evidence from the Vietnamese manufacturing sector.

- Sustainability 14 (8), 4662.
- Imran, M.T.I., Karmaker, C.L., Karim, R., Misbaudhin, S.M., Bari, A.M., Raihan, A., 2024. Modeling the supply chain sustainability imperatives in the fashion retail industry: Implications for sustainable development. *PLoS One* 19 (12), e0312671.
- Islam, S., Hassan, S., Hossain, S., Ahmed, T., Karmaker, C.L., Bari, A.M., 2025. Exploring the influence of circular economy on big data analytics and supply chain resilience nexus: a structural equation modeling approach. *Green Technol. Sustain.* 3 (3), 100219.
- Jagoda, S.U.M., Gamage, J.R., Karunathilake, H.P., 2023. Environmentally sustainable plastic food packaging: a holistic life cycle thinking approach for design decisions. *J. Clean. Prod.* 400, 136680.
- Jagtap, S., Trollman, H., Trollman, F., Garcia-Garcia, G., Parra-Lopez, C., Duong, L., Afy- Shararah, M., 2022. The Russia-Ukraine conflict: its implications for the global food supply chains. *Foods* 11 (14), 2098.
- Jain, A., Sarsaiya, S., Gong, Q., Wu, Q., Shi, J., 2024. Bioresources from biowaste for advancing manufacturing of packaging materials with modern circular economy towards a sustainable approach. *Environ., Develop. Sustain.* 1–34.
- Jerin, I., Jantan, A.H., Abdul Hamid, A.B., 2025. Intention to adopt next-level technology in food and beverage manufacturing SMEs in Bangladesh: UTAUT model and business continuity theory. *J. Inform. Technol. Manage.* 17 (1), 33–57.
- Kadam, C., 2024. Green marketing strategies in developing and developed markets.
- Kalpande, S.D., Toke, L.K., 2021. Assessment of green supply chain management practices, performance, pressure and barriers amongst Indian manufacturer to achieve sustainable development. *Int. J. Product. Perform. Manag.* 70 (8), 2237–2257.
- Kandasamy, J., Kek, V., Ethirajan, M., Murali, N., 2025. Decision making Models in Circular Supply Chain. In: *Circular Supply Chain Management: A Sustainability Approach*. Springer Nature Switzerland, Cham, pp. 121–141.
- Kannan, D., Shankar, K.M., Gholipour, P., 2022. Paving the way for a green transition through mitigation of green manufacturing challenges: a systematic literature review. *J. Clean. Prod.* 368, 132578.
- Kar, S.K., Harichandan, S., 2022. Green marketing innovation and sustainable consumption: a bibliometric analysis. *J. Clean. Prod.* 361, 132290.
- Karmaker, C.L., Bari, A.M., Anam, M.Z., Ahmed, T., Ali, S.M., de Jesus Pacheco, D.A., Moktadir, M.A., 2023. Industry 5.0 challenges for post-pandemic supply chain sustainability in an emerging economy. *Int. J. Prod. Econ.* 258, 108806.
- Karuppiah, K., Sankaranarayanan, B., Ali, S.M., 2023. A systematic review of sustainable business models: Opportunities, challenges, and future research directions. *Decision Anal.* 1, 100272.
- Karuppiah, K., Sankaranarayanan, B., Lo, H.W., 2024. A systematic literature review on the evolution of sustainable manufacturing practices: key findings and implications. *Cleaner Eng. Technol.* 22, 100798.
- Karuppiah, K., Sankaranarayanan, B., Ali, S.M., Chowdhury, P., Paul, S.K., 2020. An integrated approach to modeling the barriers in implementing green manufacturing practices in SMEs. *J. Clean. Prod.* 265, 121737.
- Kasmi, M., 2018. Biological processes as promoting way for both treatment and valorization of dairy industry effluents. *Waste Biomass Valoriz.* 9 (2), 195–209.
- Khan, I.S., Ahmad, M.O., Majava, J., 2021. Industry 4.0 and sustainable development: a systematic mapping of triple bottom line, Circular Economy and Sustainable Business Models perspectives. *J. Clean. Prod.* 297, 126655.
- Khan, M.S., Saengon, P., Alganad, A.M.N., Chongcharoen, D., Farrukh, M., 2020. Consumer green behaviour: an approach towards environmental sustainability. *Sustain. Dev.* 28 (5), 1168–1180.
- Khan, T., Emon, M.M.H., Siam, S.A.J., 2024. Impact of Green Supply Chain Practices on Sustainable Development in Bangladesh. Available at SSRN 4958443.
- Khang, A., Akhai, S., 2024. Green intelligent and sustainable manufacturing: key advancements, benefits, challenges, and applications for transforming industry. *Mach. Vision Ind. Robotics Manuf.* 405–417.
- Kizielewicz, B., Tomczyk, T., Gandor, M., Salabun, W., 2024. Subjective weight determination methods in multi-criteria decision-making: a systematic review. *Procedia Comput. Sci.* 246, 5396–5407.
- Kong, L., Wang, L., Li, F., Guo, J., 2022. Toward product green design of modeling, assessment, optimization, and tools: a comprehensive review. *Int. J. Adv. Manuf. Technol.* 122 (5), 2217–2234.
- Kuppan, N., Padman, M., Mahadeva, M., Srinivasan, S., Devarajan, R., 2024. A comprehensive review of sustainable bioremediation techniques: eco friendly solutions for waste and pollution management. *Waste Manage. Bull.*
- Kurniawan, T.A., Meidiana, C., Othman, M.H.D., Goh, H.H., Chew, K.W., 2023. Strengthening waste recycling industry in Malang (Indonesia): lessons from waste management in the era of Industry 4.0. *J. Clean. Prod.* 382, 135296.
- Kurtsal, Y., Ayalp, E.K., Viaggi, D., 2020. Exploring governance mechanisms, collaborative processes and main challenges in short food supply chains: the case of Turkey. *Bio-Based Appl. Econom.* 9 (2), 201–221.
- Kwakwa, P.A., 2022. The effect of industrialization, militarization, and government expenditure on carbon dioxide emissions in Ghana. *Environ. Sci. Pollut. Res.* 29 (56), 85229–85242.
- Laureti, L., Costantiello, A., Anobile, F., Leogrande, A., Magazzino, C., 2024. Waste Management and Innovation: insights from Europe. *Recycling* 9 (5), 82.
- Li, J., Sarkis, J., 2022. Product eco-design practice in green supply chain management: a China-global examination of research. *Nankai Bus. Rev. Int.* 13 (1), 124–153.
- Li, L., Lei, B., Mao, C., 2022. Digital twin in smart manufacturing. *J. Ind. Inf. Integr.* 26, 100289.
- Li, Y., Lin, A., 2024. Assessing the impact of green finance on financial performance in chinese eco-friendly enterprise. *Heliyon* 10 (7).
- Liu, P., Yi, S.P., 2017. Pricing policies of green supply chain considering targeted advertising and product green degree in the big data environment. *J. Clean. Prod.* 164, 1614–1622.
- Luthra, S., Sharma, M., Kumar, A., Joshi, S., Collins, E., Mangla, S., 2022. Overcoming barriers to cross-sector collaboration in circular supply chain management: a multi-method approach. *Transport. Res. Part e: Logistics Transport. Rev.* 157, 102582.
- Maldonado-Romo, J., Ahmad, R., Ponce, P., Mendez, J.I., Mata, O., Rojas, M., Molina, A., 2024. Advancing sustainable manufacturing: a case study on plastic recycling. *Prod. Manuf. Res.* 12 (1), 2425672.
- Mason, A., Haidegger, T., Alvsøike, O., 2023. Time for change: the case of robotic food processing [industry activities]. *IEEE Rob. Autom. Mag.* 30 (2), 116–122.
- Matos, J.D.P., Rodrigues, M.B., Duarte, C.K., Horta, P.M., 2023. A scoping review of observational studies on food and beverage advertising on social media: a public health perspective. *Int. J. Environ. Res. Public Health* 20 (4), 3615.
- Memon, S.U.R., Pawase, V.R., Pavase, T.R., Soomro, M.A., 2021. Investigation of COVID- 19 impact on the food and beverages industry: China and India perspective. *Foods* 10 (5), 1069.
- Mer, A., Virdi, A.S., 2024. Decoding the challenges and skill gaps in small-and medium- sized enterprises in emerging economies: a review and research agenda. *Contemporary Challenges in Social Science Management: Skills Gaps and Shortages in the Labour Market* 112, 115–134.
- Michel, M., Eldridge, A.L., Hartmann, C., Klassen, P., Ingram, J., Meijer, G.W., 2024. Benefits and challenges of food processing in the context of food systems, value chains and sustainable development goals. *Trends Food Sci. Technol.* 104703.
- Modares, A., Emroozi, V.B., Roozkhosh, P., Modares, A., 2025. A Bayesian best-worst approach with blockchain integration for optimizing supply chain efficiency through supplier selection. *Supply Chain Anal.* 9, 100100.
- Mohammadi, M., Rezaei, J., 2020. Bayesian best-worst method: a probabilistic group decision making model. *Omega* 96, 102075.
- Mohideen, M.M., Ramakrishna, S., Prabhu, S., Liu, Y., 2021. Advancing green energy solution with the impetus of COVID-19 pandemic. *J. Energy Chem.* 59, 688–705.
- Mollah, A.H., Wahab, M., Srabon, S.S., 2024. The effect of sustainable packaging practices on youthful bangladeshi consumers' preferences and purchasing behaviour: application on food and beverage market. *South Asian J. Manag.* 18 (1), 20–44.
- Mongabay. Bangladesh apparel industry makes progress in 'eco-friendly' manufacturing. Available online: <https://news.mongabay.com/2023/09/bangladesh-apparel-indus-try-makes-progress-in-eco-friendly-manufacturing/> (accessed on 19 September 2023).
- Morshed, S.M., Shakur, M.S., Rahman, R., Mahin, M.M.I., Debnath, B., Apu, A.H., Bari, A. M., 2025. An interval-valued type 2 intuitionistic fuzzy theory-based approach to assess the biofuel production and adoption drivers in emerging economies: Implications for sustainability. *Energy Nexus* 17, 100369.
- Munim, Z.H., Chowdhury, M.M.H., Tusher, H.M., Notteboom, T., 2023. Towards a prioritization of alternative energy sources for sustainable shipping. *Mar. Policy* 152, 105579.
- Munim, Z.H., Mohammadi, M., Shakil, M.H., Ali, S.M., 2022. Assessing measures implemented by export-oriented RMG firms in an emerging economy during COVID- 19. *Comput. Ind. Eng.* 165, 107963.
- Mykoniatis, K., Harris, G.A., 2021. A digital twin emulator of a modular production system using a data-driven hybrid modeling and simulation approach. *J. Intell. Manuf.* 32 (7), 1899–1911.
- Nagaj, R., Gajdzik, B., Wolniak, R., Grebski, W.W., 2024. The impact of deep decarbonization policy on the level of greenhouse gas emissions in the European Union. *Energies* 17 (5), 1245.
- Nazir, M., Arif, S., Khan, R.S., Nazir, W., Khalid, N., Maqsood, S., 2019. Opportunities and challenges for functional and medicinal beverages: current and future trends. *Trends Food Sci. Technol.* 88, 513–526.
- Nazir, S., Zhaolei, L., Mehmood, S., Nazir, Z., 2024. Impact of green supply chain management practices on the environmental performance of manufacturing

- firms considering institutional pressure as a moderator. *Sustainability* 16 (6), 2278.
- Ndzabukelwako, Z., Mereko, O., Sambo, T., Thango, B., 2024. The Impact of Porter's Five Forces Model on SMEs Performance: A Systematic Review. Available at SSRN 499059.
- Nguyen, H.T., Hoang, T.G., Nguyen, L.Q.T., Le, H.P., Mai, H.X.V., 2022. Green technology transfer in a developing country: mainstream practitioner views. *Int. J. Organ. Anal.* 30 (3), 699–720.
- Ni'zeti'c, S., Djilali, N., Papadopoulos, A., Rodrigues, J.J., 2019. Smart technologies for promotion of energy efficiency, utilization of sustainable resources and waste management. *J. Clean. Prod.* 231, 565–591.
- Odoro, S., Maccario, G., De Nisco, A., 2022. Green innovation: a multidomain systematic review. *Eur. J. Innov. Manag.* 25 (2), 567–591.
- Ojiako, U., Bashir, H., Almansoori, H.H.B., AlRaeesi, E.J.H., Al Zarooni, H.A., 2024. Using ISO 14001 certification to signal sustainability equivalence: an examination of the critical success factors. *Prod. Plan. Control* 1–28.
- Ojo, L.D., Oladimirin, O.T., Chan, A.P., Owusu, E.K., 2024. Understanding the fundamental barriers to environmental management system implementation in developing countries. *J. Build. Des. Environ.* 3 (3), 36971.
- Osman, A.I., Chen, L., Yang, M., Msigwa, G., Farghali, M., Fawzy, S., Yap, P.S., 2023. Cost, environmental impact, and resilience of renewable energy under a changing climate: a review. *Environ. Chem. Lett.* 21 (2), 741–764.
- P Shrivani, K., Vaishali, S., 2024. Identifying and disseminating ESG best practices: route map to strengthening sustainability ecosystem. *Int. J. Trend Sci. Res. Develop.* 8 (3), 72–76.
- Petrescu, M.G., Panait, M., Fu, H., 2021. Integrated management systems under the banner of sustainable development: risks and opportunities. *Sustain. Manage. Managers Eng.* 157–187.
- Phonthanukitithaworn, C., Srisathan, W.A., Naruetharadhol, P., 2024. Revolutionizing waste management: Harnessing citizen-driven innovators through open innovation to enhance the 5Rs of circular economy. *J. Open Innovat.: Technol., Market, Complexity* 10 (3), 100342.
- Pimenov, D.Y., Mía, M., Gupta, M.K., Machado, A.R., Pintaude, G., Unune, D.R., Kuntoglu, M., 2022. Resource saving by optimization and machining environments for sustainable manufacturing: a review and future prospects. *Renew. Sustain. Energy Rev.* 166, 112660.
- Poore, J., Nemecek, T., 2018. Reducing food's environmental impacts through producers and consumers. *Science* 360 (6392), 987–992.
- Prasanna, S., Verma, P., Bodh, S., 2024. The role of food industries in sustainability transition: a review. *Environ. Develop. Sustain.* 1–21.
- Premadasa, M., Siyambalapitiya, J., Jayawardhana, K., Fernando, I., 2023. Conceptualizing the role of social entrepreneurial orientation in the triple bottom line in the social enterprise context: developing country perspective. *Sustainability* 15 (11), 8759.
- Queiroz, G.A., Filho, A.G.A., Núñez, J.F., Santa-Eulalia, L.A., Delai, I., Torkomian, A.L.V., 2024. Lean and Green Manufacturing in operations strategy: cases from the automotive industry. *Oper. Manag. Res.* 1–25.
- Rahman, M., Emon, M.E.H., Antor, M.H., Haque, S.A., Talapatra, S., 2025. Identification and prioritization of barriers to Industry 4.0 adoption in the context of food and beverage industries of Bangladesh. *BIJ* 32 (2), 757–783.
- Raja, K.S., Maliha, A., Paul, S., Shakur, M.S., Bari, A.M., 2025. Unraveling the obstacles to transform agro-wastes into value-added bioproducts: implications for sustainability. *Sustainable Oper. Comput.* <https://doi.org/10.1016/j.susoc.2025.05.002>.
- Rajpoot, S., 2024. Eco-Friendly Solutions for Sustainable Waste Management: an Approach Towards a Cleaner and Greener Future. In: *Waste Management and Life Cycle Assessment for Sustainable Business Practice*. IGI Global, pp. 43–64.
- Ramadhan, M.L., Nugraha, F., Prastowo, D.A., Kusumawardhani, A., Raharjo, S.T., 2024. Development of Environmentally Friendly Technology for Key Industries in Achieving Golden Indonesia. *Research Horizon* 4 (4), 205–220.
- Renita, A.A., Gajaria, T.K., Sathish, S., Kumar, J.A., Lakshmi, D.S., Kujawa, J., Kujawski, W., 2023. Progress and prospective of the industrial development and applications of ECO-friendly colorants: an insight into environmental impact and sustainability issues. *Foods* 12 (7), 1521.
- Rosenboom, J.G., Langer, R., Traverso, G., 2022. Bioplastics for a circular economy. *Nat. Rev. Mater.* 7 (2), 117–137.
- Runtuk, J.K., Ng, P.K., Ooi, S.Y., 2024. Challenges and Solutions in Working with Green suppliers: Perspective from a Manufacturing Industry. *Sustainability* 16 (20), 8744.
- Sabet, M., 2024. Exploring biodegradable polymer composites for sustainable packaging: a review on properties, manufacturing techniques, and environmental impacts. *Iran. Polym. J.* 1–20.
- Sabourin, D., 2021. *Sustainability Consideration in Capital Investment Decisions: a Survey of executives and Associates Directly involved with Project Investment Approvals* (Doctoral dissertation). Colorado Technical University.
- Saleem, A., Sun, H., Aslam, J., Kim, Y., 2024. Impact of smart factory adoption on manufacturing performance and sustainability: an empirical analysis. *Bus. Process. Manag. J.*
- Salim, H.K., Padfield, R., Lee, C.T., Sayuti, K., Papargyropoulou, E., Tham, M.H., 2018. An investigation of the drivers, barriers, and incentives for environmental management systems in the Malaysian food and beverage industry. *Clean Techn. Environ. Policy* 20, 529–538.
- Saner, H.S., Yucesan, M., Gul, M., 2022. A Bayesian BWM and VIKOR-based model for assessing hospital preparedness in the face of disasters. *Nat. Hazards* 1–33.
- Sarker, A., Ghosh, M.K., Islam, T., Bilal, M., Nandi, R., Raihan, M.L., Kim, J.E., 2022. Sustainable food waste recycling for the circular economy in developing countries, with special reference to Bangladesh. *Sustainability* 14 (19), 12035.
- Savale, T.K., Byram Anand, P., Perumalla Varalaxmi, A.B., Quaye, J.A., 2023. Green marketing strategies: assessing consumer perception and adoption of eco-friendly products. *Remittances Rev.* 8 (4).
- Savvakis, N., Sifakis, N., Kotakidis, X., Arampatzis, G., 2023. Multiple energy resources integration in the food industry: a technoeconomic analysis. *J. Clean. Prod.* 426, 139055.
- Saxena, A., Srivastava, A., 2022. Industry application of green manufacturing: a critical review. *J. Sustain. Environ. Manage.* 1 (1), 32–45.
- Sembiring, R.J., 2021. The effect green marketing mix on corporate image as well as implication for purchase intention of food and beverages companies in Indonesia. *J. Soc. Sci.* 2 (2), 210–222.
- Shah, H.H., Amin, M., Pepe, F., Tregambi, C., 2024. Sustainable waste management and waste-to-energy in the context of a circular economy through various waste management technologies. *Environ. Sci. Pollut. Res.* 1–20.
- Shahin, V., Alimohammadlou, M., Abbasi, A., 2024. Identifying and prioritizing the barriers to green innovation in SMEs and the strategies to counteract the barriers: an interval-valued intuitionistic fuzzy approach. *Technol. Forecast. Soc. Chang.* 204, 123408.
- Shahzad, M., Qu, Y., Ur Rehman, S., Ding, X., Razaq, A., 2023. Impact of stakeholders' pressure on green management practices of manufacturing organizations under the mediation of organizational motives. *J. Environ. Plan. Manag.* 66 (10), 2171–2194.
- Shakur, M.S., Lubaba, M., Debnath, B., Bari, A.M., Rahman, M.A., 2024. Exploring the challenges of Industry 4.0 Adoption in the FMCG Sector: Implications for Resilient Supply Chain in Emerging Economy. *Logistics* 8 (1), 27.
- Sharma, S., Prakash, G., Kumar, A., Mussada, E.K., Antony, J., Luthra, S., 2021. Analysing the relationship of adaption of green culture, innovation, green performance for achieving sustainability: mediating role of employee commitment. *J. Clean. Prod.* 303, 127039.
- Shi, C., Guo, N., Gao, X., Wu, F., 2022. How carbon emission reduction is going to affect urban resilience. *J. Clean. Prod.* 372, 133737.
- Shukla, G.P., Swarnakar, V., Singh, S.J., 2023. Assessment of drivers and barriers of green manufacturing practices in Indian manufacturing companies. *J. Institution Engineers (India): Series C* 104 (1), 45–54.
- Sindhvani, R., Mittal, V.K., Singh, P.L., Aggarwal, A., Gautam, N., 2019. Modelling and analysis of barriers affecting the implementation of lean green agile manufacturing system (LGAMS). *BIJ* 26 (2), 498–529.
- Singh, B.J., Chakraborty, A., Sehgal, R., 2023a. A systematic review of industrial wastewater management: evaluating challenges and enablers. *J. Environ. Manage.* 348, 119230.
- Singh, P.K., Chirade, S., Taluja, R., Yadav, D.K., Srikanth, A., Manjunatha, M., Karim, M. M., 2023b. Eco-Conscious Creation: Navigating the Nexus of Sustainability and Production Design Vol. 453, 01034.
- Singhal, T.S., Jain, J.K., Ramacharyulu, D.A., Jain, A., Abdul-Zahra, D.S., Manjunatha, M., Srivastava, A.P., 2024. Eco-Design of Products and Processes Vol. 505, 01033.
- Sommerville, M., 2024. The Geopolitics of Food Security. In: *The Palgrave Handbook of Contemporary Geopolitics*. Springer Nature Switzerland, Cham, pp. 1–31.
- Sovacool, B.K., Bazilian, M., Griffiths, S., Kim, J., Foley, A., Rooney, D., 2021. Decarbonizing the food and beverages industry: a critical and systematic review of developments, sociotechnical systems and policy options. *Renew. Sustain. Energy Rev.* 143, 110856.
- Srisathan, W.A., Ketkaew, C., Phonthanukitithaworn, C., Naruetharadhol, P., 2023. Driving policy support for open eco-innovation enterprises in Thailand: a probit regression model. *J. Open Innovation: Technol., Market, Complexity* 9 (3), 100084.
- Sukrana, B., Hassan, S., Jui, F.I., Shakur, M.S., Debnath, B., Bari, A.B.M., 2025. Investigating the influence of the push and pull factors in eco-resort selection to promote sustainable Tourism in Bangladesh. *Sustainable Futures* 9, 100619.
- Taghipour, A., Akkatham, W., Eaknarajindawat, N., Stefanakis, A.I., 2022. The impact of government policies and steel recycling companies' performance on sustainable management in a circular economy. *Resour. Policy* 77, 102663.
- Takacs, F., Brunner, D., Frankenberger, K., 2022. Barriers to a circular economy in small- and medium-sized enterprises and their integration in a sustainable strategic management framework. *J. Clean. Prod.* 362, 132227.

- Tyagi, A.K., Bhatt, P., Chidambaram, N., Kumari, S., 2024. Artificial intelligence empowered smart manufacturing for modern society: a review. *Artif. Intell.-Enabled Digital Twin for Smart Manuf.* 55–83.
- Ullah, S., Ahmad, N., Khan, F.U., Badulescu, A., Badulescu, D., 2021. Mapping interactions among green innovations barriers in manufacturing industry using hybrid methodology: insights from a developing country. *Int. J. Environ. Res. Public Health* 18 (15), 7885.
- Ullah, S., Khan, F.U., Ahmad, N., 2022. Promoting sustainability through green innovation adoption: a case of manufacturing industry. *Environ. Sci. Pollut. Res.* 29 (14), 21119–21139.
- Umar, M., Ahmad, A., Sroufe, R., Muhammad, Z., 2024. The nexus between green intellectual capital, blockchain technology, green manufacturing, and sustainable performance. *Environ. Sci. Pollut. Res.* 31 (10), 15026–15038.
- Unilever Bangladesh. Unilever Bangladesh Recognised for Leadership in Sustainability. Available online: <https://www.unilever.com.bd/news/2024/unilever-bangladesh-re-ognised-for-leadership-in-sustainability/> (accessed on 22 October 2024).
- Usman, F.O., Ani, E.C., Ebirim, W., Montero, D.J.P., Olu-lawal, K.A., Ninduwezuor-Ehiobu, N., 2024. Integrating renewable energy solutions in the manufacturing industry: challenges and opportunities: a review. *Eng. Sci. Technol. J.* 5 (3), 674–703.
- Vangeri, A.K., Bathrinath, S., Anand, M.C.J., Shanmugathai, M., Meenatchi, N., Boopathi, S., 2024. Green Supply Chain Management in Eco-Friendly Sustainable Manufacturing Industries. In: *Environmental Applications of Carbon-Based Materials*, IGI Global, pp. 253–287.
- Virmani, N., Bera, S., Kumar, R., 2021. Identification and testing of barriers to sustainable manufacturing in the automobile industry: a focus on Indian MSMEs. *BIJ* 28 (3), 857–880.
- Wang, J., Azam, W., 2024. Natural resource scarcity, fossil fuel energy consumption, and total greenhouse gas emissions in top emitting countries. *Geosci. Front.* 15 (2), 101757.
- Wang, W., Zhang, D., Wang, H., Zhu, Q., Morabbi Heravi, H., 2023. How do businesses achieve sustainable success and gain a competitive advantage in the green era? *Kybernetes* 52 (9), 3241–3260.
- World Bank. Green Growth Framework Can Help Bangladesh Achieve Sustainable Development. Available online: <https://www.worldbank.org/en/news/press-release/2024/06/13/green-growth-framework-can-help-bangladesh-achieve-sustainable-development> (accessed on 13 June, 2024).
- World Bank, 2021. Sustainable Plastic Management is key to achieve Green Growth for Bangladesh. Available online: <https://www.worldbank.org/en/news/press-release/2021/12/20/sustainable-plastic-management-is-key-to-achieve-green-growth-for-bangladesh> (accessed on 20 December, 2021).
- Xu, L., Guo, P., Wen, H., 2023. Increasing short-term lending for long-term investment under environmental pressure: evidence from China's energy-intensive firms. *Environ. Sci. Pollut. Res.* 30 (6), 14693–14706.
- Yang, Y., Tilman, D., Jin, Z., Smith, P., Barrett, C.B., Zhu, Y.G., Zhuang, M., 2024. Climate change exacerbates the environmental impacts of agriculture. *Science* 385 (6713), eadn3747.
- Yang, Z., Shang, W.L., Zhang, H., Garg, H., Han, C., 2022. Assessing the green distribution transformer manufacturing process using a cloud-based q-rung orthopair fuzzy multi-criteria framework. *Appl. Energy* 311, 118687.
- Yuan, B., Cao, X., 2022. Do corporate social responsibility practices contribute to green innovation? The mediating role of green dynamic capability. *Technol. Soc.* 68, 101868.
- Zain, R.M., Ramli, A., Zain, M.Z.M., Yekini, L.S., Musa, A., Rahim, M.N.A., Aziz, N.I.C., 2024. An Investigation of the barriers and drivers for implementing green supply chain in Malaysian food and beverage SMEs: a qualitative perspective. *WSEAS Trans. Bus. Econ.* 21, 2169–2189.
- Zhang, D., Vigne, S.A., 2021. How does innovation efficiency contribute to green productivity? A financial constraint perspective. *J. Clean. Prod.* 280, 124000.
- Zheng, Y., Wu, Y., Zhang, Y., Meng, X., Zhang, P., 2025. Greening the future: how green manufacturing shapes corporate environmental and ESG success. *Int. Rev. Financ. Anal.*
- Zim, Z.S., Zahan, M., 2019. Local versus global brands: the internationalization process in Food and Beverage Industry. *Business Management Rev.* 10 (5), 58–66.
- Zorpas, A.A., 2024. The hidden concept and the beauty of multiple “R” in the framework of waste strategies development reflecting to circular economy principles. *Sci. Total Environ.* 952, 175508.